

CHEM 1043
General Chemistry II

2026-2027

Semester 1

Course Information:

Course website:	Moodle course *(access via Student Portal → My eLearning)*
No. of credits:	6
Pre-requisite:	Pass in CHEM1042 General Chemistry I
Duration:	Semester 1
Teaching:	36 hours of lectures plus ~10 hours of tutorials
Course Coordinator:	Dr. Arnold Li (litla@hku.hk)
Teachers:	Prof. Seungkyu Lee (Topic 1-3) (skchem@hku.hk, Rm 302, CYM Chemistry Building, office hour: Tue 10-11 am)
Assessment:	Exam (50 %), Test (20 %) and two assignments (15 % each)

Teaching Assistants for CHEM1043

Mr. Zheqi LU (zheqilu_chem@connect.hku.hk)	Responsible for Topics 1 & 2
Ms. Chenxi XIONG (xiongchx@connect.hku.hk)	Responsible for Topics 3 & 4
Mr. Mengtian ZHANG (skydreamerzmt@connect.hku.hk)	Responsible for Topics 5 & 6

Please refer to the document “CHEM1043 Tutorial & Consultation Schedule” for more details.

Your Prior Knowledge in Chemistry



CHEM1042 General Chemistry I

(1) Chemistry: its nature and method

Physical properties; chemical changes and chemical properties; elements and compounds; measuring mass, length, volume and temperature; atomic structure and subatomic particles; the mole concept and stoichiometry; solutions and concentrations; uncertainty in measurement and significant figures.

(2) Atoms: the quantum world

Electromagnetic radiation and matter; Planck's quantum theory; the Bohr model of the hydrogen atom; the quantum mechanical model of the atom; quantum numbers, energy levels, and atomic orbitals; shapes of atomic orbitals; electron configurations; periodic trends: atomic radii, ionic radii, ionization energies, and electron affinities.

(3) Chemical bonding and structures

Review on covalent, ionic and metallic bond. Covalent bonds and molecular structures (VSEPR, VB theory).

(4) Energetics and kinetics of reactions

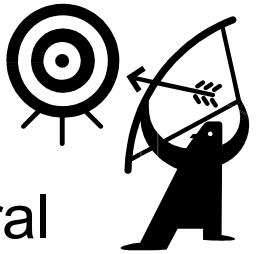
Heat and work; the first law of thermodynamics; heat of reactions; spontaneity of changes.

Reaction rate; factors that influence reaction rate; rate laws: differential and integrated rate laws; temperature and reaction rate; reaction mechanisms.

(5) Acid-Base equilibria

Acid-base concepts; equilibria in solutions of weak acids and in weak bases; ionization constants; molecular properties and acid strength; acid-base properties of salt solutions; buffer solutions; acid-base titrations.

Course Objectives:



- This course is a continuation of CHEM1042 General Chemistry I. It aims to further consolidate some of the important fundamentals of chemistry that underlie many topics and principles across the physical sciences.
- The course prepares students to pursue a major in chemistry or in other aspects that require a good foundation in chemistry.

Course Contents:

This content list is subject to minor change.

(1) Gases

Simple gas laws; ideal gas equation; gases in chemical reactions; mixture of gases; kinetic-molecular theory of gases; diffusion and effusion; non-ideal gases.

(2) Structure and Bonding: The Delocalized Approach: Molecular Orbital Theory

Bonding in homonuclear and heteronuclear diatomic molecules of first and second period of elements; bonding in some simple polyatomic molecules; bonding in metals (band theory).

(3) Solutions and their Properties

Types of solutions; intermolecular forces and the solution process; solution formation and equilibrium; solubilities of gases; vapor pressures of solutions; osmotic pressure; freezing-point depression and boiling-point elevation of nonelectrolyte solutions; solutions of electrolytes; colloidal mixtures.

(4) Solubility and Complex-Ion Equilibria

Solubility product constant; relationship between solubility and K_{sp}; common-ion effect in solubility equilibria; limitations of the K_{sp} concept; precipitation; solubility and pH; equilibria involving complex ions; qualitative cation analysis.

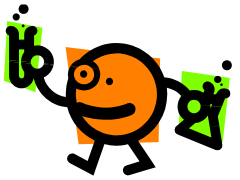
(5) Entropy & Gibbs Energy

A quick review on entropy and the second & third laws of thermodynamics. Standard Gibbs energy change; Gibbs energy change and equilibrium; coupled reactions.

(6) Electrochemistry

Electrode potentials and their measurement; standard electrode potentials; E_{cell}, ΔG, and K; E_{cell} as a functions of concentrations; batteries; corrosion; electrolysis; industrial electrolysis processes.

Course Learning Outcomes:



On successful completion of this course, students should be able to:

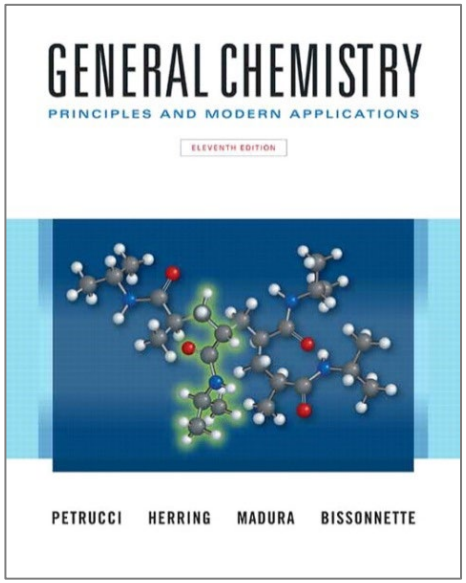
1. Demonstrate a knowledge and understanding of the properties and behavior of gases and apply gas laws and kinetic-molecular theory to processes involving gases.
2. Demonstrate a knowledge and understanding in relation to solutions and their properties, solubility and complex-ion equilibria, and also electrochemistry.
3. Apply molecular orbital theory to explain the formation and properties of diatomic molecules of first and second period of elements and of some simple polyatomic molecules.
4. Demonstrate a knowledge and understanding of the relationship between free energy and spontaneity of reaction.
5. Apply the theories and concepts introduced in the course to solve problems, perform calculations, make predictions and rationalize trends.
6. Organize and present chemical ideas in a clear, logical and coherent way.
7. Demonstrate awareness of the relevant applications of chemistry in society and in everyday life.

Textbook & Reference Books:

Reading is important!

Textbook:

Petrucci; Herring; Madura; Bissonnette: *General Chemistry: Principles and Modern Applications*, 11th edition, Pearson



Other reference books:

1) Moore; Stanitski; Jurs: *Chemistry: The Molecular Science*, latest edition, Brookes/Cole

2) Zumdahl; Zumdahl: *Chemistry*, latest edition, Brookes/Cole

3) Silberberg: *Chemistry: The Molecular Nature of Matter and Change*, latest edition, McGraw Hill

4) Raymond Chang: *Chemistry*, latest edition, McGraw Hill

Purchase link:
https://apbookshop.com/HKU/EN/Display/PrintedBook?ISBN=9780134097329



CHEM 1043 General Chemistry II

Topic 1 Gases: Properties and Behaviour

Prof. Seungkyu Lee

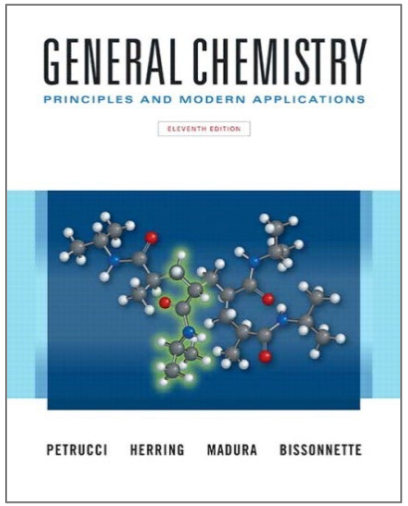


Your reading:

Chapter 6) Gases

Petrucci; Herring; Madura; Bissonnette:

General Chemistry: Principles and Modern Applications, 11th edition, Pearson





Why study gases?

- E.g.) The earth's atmosphere is ***a mixture of gases*** that consists mainly of nitrogen (N_2), oxygen (O_2), and carbon dioxide (CO_2) molecules. The atmosphere supports life.
- E.g.) How can we remove CO_2 from the atmosphere?
- E.g.) Many industrial sectors are trying to develop safe and efficient H_2-storage systems for the hydrogen economy.
- E.g.) Gasoline (gas) reacts with O_2 and releases energy in engines.



Why study gases?

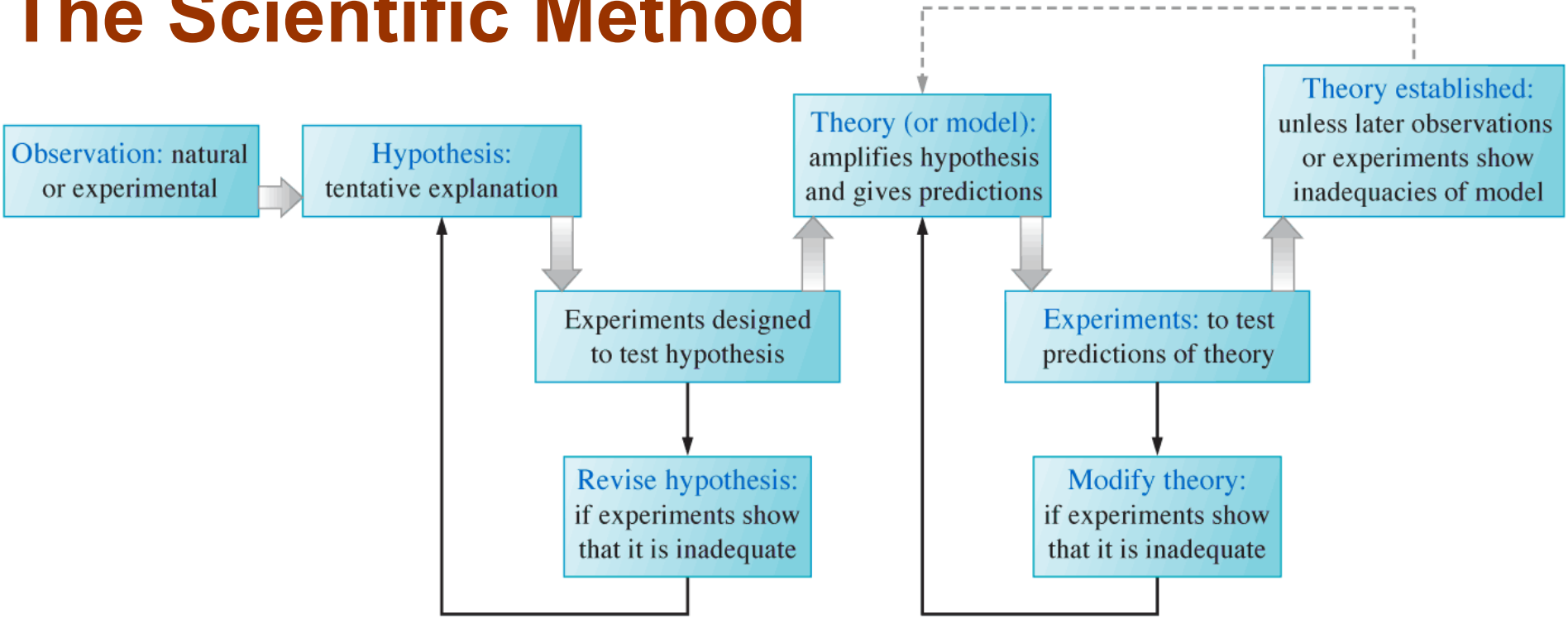
- Although relatively few substances exist in the gaseous state under typical condition, gases are very important.

- E.g.) the earth's atmosphere is ***a mixture of gases*** that consists mainly of elemental nitrogen (N_2) and oxygen (O_2). The atmosphere supports life.

Elements that exist as *gases* at 25°C and 1 atmosphere

1A												3A	4A	5A	6A	7A	8A
H	2A																He
Li	Be											B	C	N	O	F	Ne
Na	Mg	3B	4B	5B	6B	7B		8B			1B	2B	Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr	
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe	
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn	
Fr	Ra	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg								

How are we going to study gases?
The Scientific Method



e.g. Observation → Natural Law (gas laws) → Model/Theory (kinetic-molecular theory of gases)

Scientific laws are statements, based on repeated experiments or observations, that describe or predict a range of natural phenomena

What to learn in this topic?



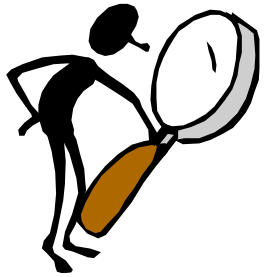
1. Properties of gases
2. The simple gas laws
3. The ideal gas equation & general gas equation
4. Reaction stoichiometry
5. Mixture of gases
6. Kinetic-molecular theory of gases
7. Gas properties relating to the kinetic-molecular theory
8. Non-ideal (real) gases

TABLE 5.1 Some Substances Found as Gases at 1 atm and 25°C

Elements	Compounds
H_2 (molecular hydrogen)	HF (hydrogen fluoride)
N_2 (molecular nitrogen)	HCl (hydrogen chloride)
O_2 (molecular oxygen)	HBr (hydrogen bromide)
O_3 (ozone)	HI (hydrogen iodide)
F_2 (molecular fluorine)	CO (carbon monoxide)
Cl_2 (molecular chlorine)	CO_2 (carbon dioxide)
He (helium)	NH_3 (ammonia)
Ne (neon)	NO (nitric oxide)
Ar (argon)	NO_2 (nitrogen dioxide)
Kr (krypton)	N_2O (nitrous oxide)
Xe (xenon)	SO_2 (sulfur dioxide)
Rn (radon)	H_2S (hydrogen sulfide)
	HCN (hydrogen cyanide)*

*The boiling point of HCN is 26°C, but it is close enough to qualify as a gas at ordinary atmospheric conditions.

In this topic,



We will look at

- the properties of gases

- how the bahaviour of the atoms and molecules that are the basic components of gases determine the properties of gases.

How to proceed?

- We will proceed from the macroscopic to the microscopic.
- To examine the properties of gases, such as pressure (P), volume (V), temperature (T), and amount of gas (n) (the simple gas laws and the ideal gas equation).
- These laws will be explained by the kinetic-molecular theory of gases.

1. Properties of Gases

- Gases expand to fill and assume the shapes of their containers.
- They diffuse into one another and mix in all proportions.
- Gases are the most *compressible* state of matter.
- Gases have much lower densities than liquids and solids.
- ***Four properties*** determine the physical behaviour of a gas:
  1. the amount of the gas (i.e. no. of moles) **n**
  2. the volume of the gas **V**
  3. the temperature of the gas **T**
  4. the pressure of the gas **P**



(Left) Chlorine gas (greenish yellow), (Centre) Bromine gas (brownish red) above liquid bromine, (Right) Iodine gas (violet) above solid iodine.

The Concept of Pressure



- *Pressure* is defined as a force per unit area.

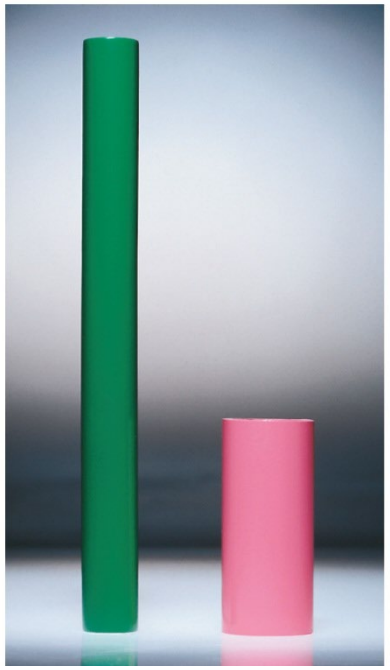
$$P\ (Pa) = \frac{\text{Force (N)}}{\text{Area } (m^2)}$$

1 Pa = 1 Nm^{-2} = 1 kg $m^{-1}s^{-2}$

The two cylinders have the same mass and exert the same force on the supporting surface.

$$\text{Force (N)} = m\ (kg) \times g\ (m/s^2)$$

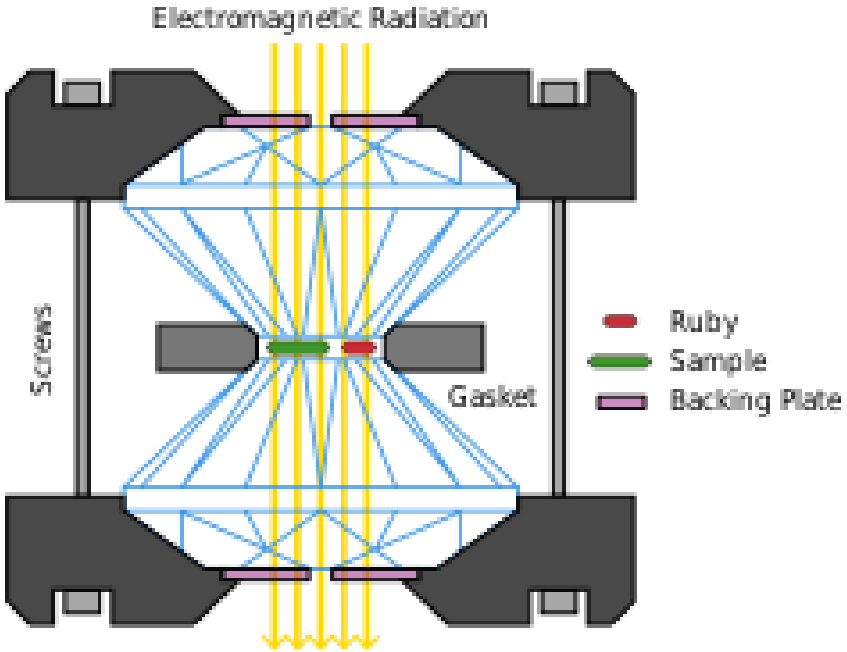
Which one exerts a greater pressure?

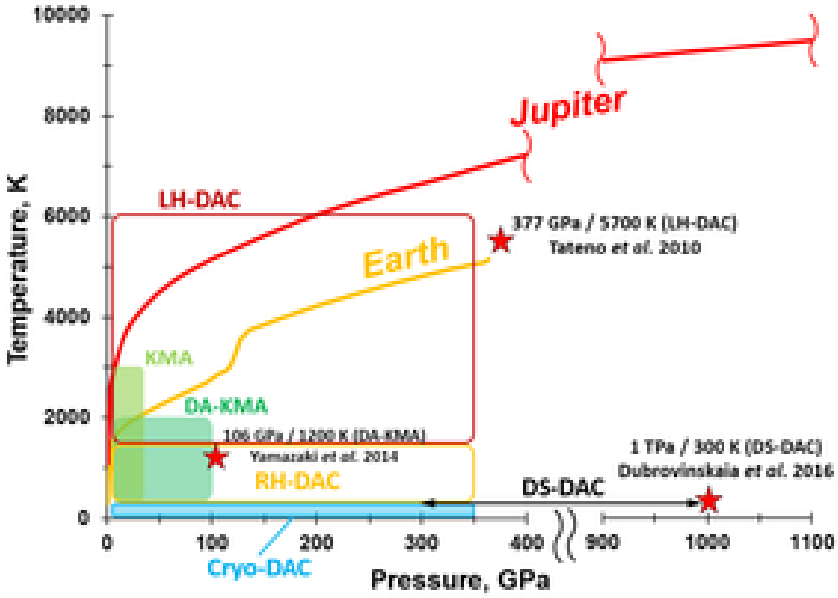


Can we create ultra-high pressures similar to those in the core of the Earth in the lab?

$$P\ (Pa) = \frac{\text{Force (N)}}{\text{Area } (m^2)}$$

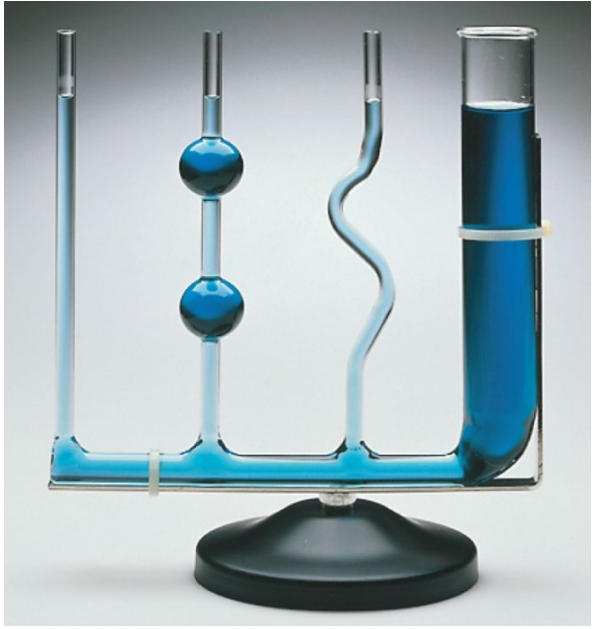
Diamon Anvil cell (DAC)





Liquid Pressure

- Long time ago, the pressure of a gas is measured indirectly, by comparing it with a liquid pressure.
- The pressure of a liquid depends only on the height (h) of the liquid column and the density (d) of the liquid.



All the interconnected vessels fill to the same height. As a result, the liquid pressures are the same despite the different shapes and volumes of the containers.

W = weight, m = mass, g = gravity, V = volume

$$P\,(Pa) = \frac{F}{A} = \frac{W}{A} = \frac{m \times g}{A} = \frac{V \times d \times g}{A} = \frac{h \times A \times d \times g}{A} = h \times d \times g$$

weight is force

$W = m \times g$

$m = V \times d$

$V = A \times h$

g is constant

Conclusion:

Liquid pressure ∝ liquid density and the height of the liquid column

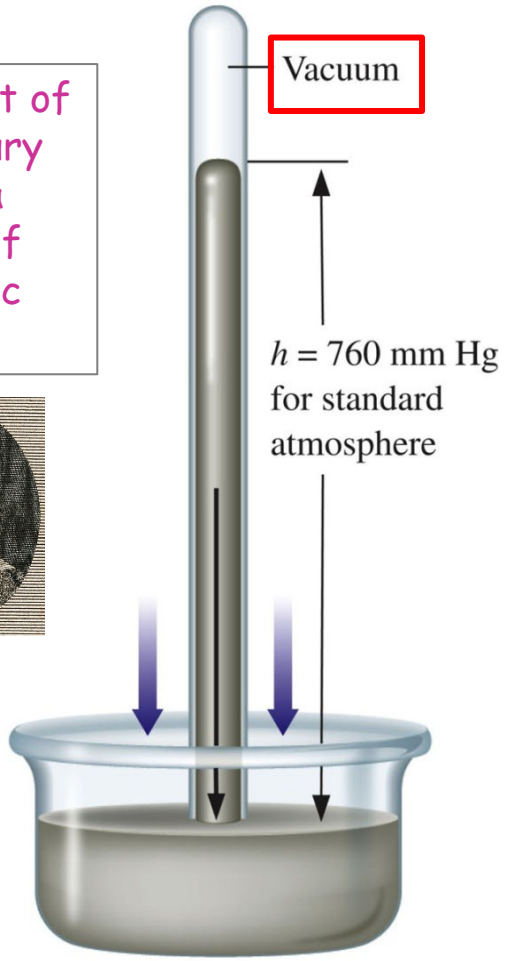
Barometric Pressure

- **Barometer**, is a device constructed by Evangelista Torricelli to measure **atmospheric pressure (1643)**.

- Mercury flows out of the tube until the pressure of the column of mercury standing on the surface of the mercury in the dish is *equal* to the pressure of the air on the rest of the surface of the mercury in the dish.

The height of the mercury provides a measure of barometric pressure.





Standard Atmospheric Pressure

1 atm = 101.325 kPa = 1.01325 bar = 760 Torr ≈ 760 mmHg

Units of Pressure

Some Common Pressure Units

Atmosphere	atm	
Millimeter of mercury	mmHg	1 atm ≃ 760 mmHg
Torr	Torr	= 760 Torr
Pascal	Pa	= 101,325 Pa
Kilopascal	kPa	= 101.325 kPa
Bar	bar	= 1.01325 bar
Millibar	mbar	= 1013.25 mbar

1 bar = 100,000 Pa

1 atm ≈ 1 bar

Standard Conditions of Temperature and Pressure

- Gas properties depend on conditions.
- It is useful to have a set of standard conditions of temperature and pressure that can be used for comparing different gases.
- IUPAC defines standard conditions of temperature and pressure (STP) as follows:

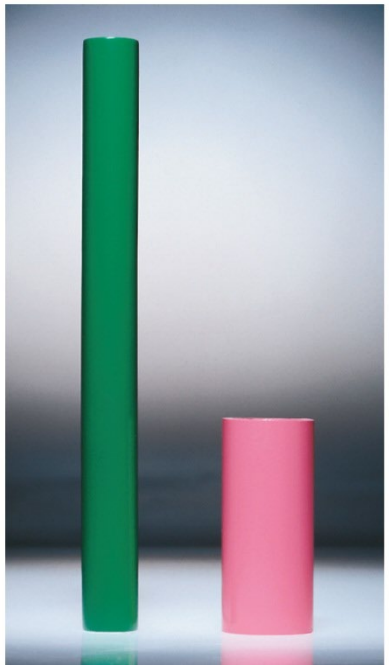
$$P = 1 \text{ bar} = 10^5 \text{ Pa}$$

$$T = 0 \text{ °C} = 273.15 \text{ K}$$

Class Practice 3

EXAMPLE 6-3 Using SI Units of Pressure

The 1.000 kg red cylinder in Figure 6-2 has a diameter of 4.10 cm. What pressure, expressed in Torr, does this cylinder exert on the surface beneath it?



green red

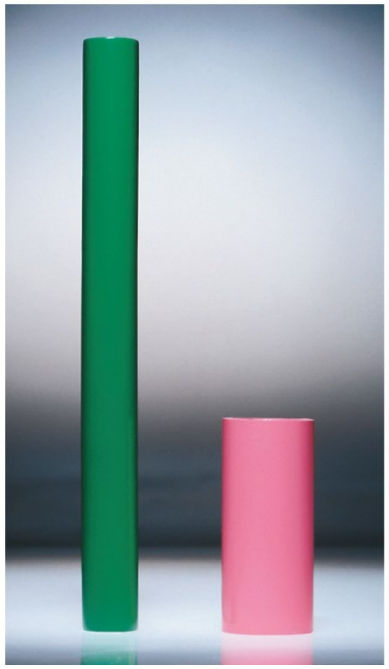
$g = 9.81\ ms^{-2}$

Atmosphere	atm	
Millimeter of mercury	mmHg	1 atm ≃ 760 mmHg
Torr	Torr	= 760 Torr
Pascal	Pa	= 101,325 Pa
Kilopascal	kPa	= 101.325 kPa
Bar	bar	= 1.01325 bar
Millibar	mbar	= 1013.25 mbar

Class Practice 3

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green red

$g = 9.81\ ms^{-2}$

$$P\,(Pa) = \frac{F}{A} = \frac{W}{A} = \frac{m \times g}{A} = \frac{V \times d \times g}{A} = \frac{h \times A \times d \times g}{A}$$

$$= h \times d \times g$$

Class Practice 3 (Con't)

$$P\,(Pa) = \frac{F}{A} = \frac{W}{A} = \frac{m \times g}{A} = \frac{V \times d \times g}{A} = \frac{h \times A \times d \times g}{A} = h \times d \times g$$

Analyze

We must apply equation (6.2). It is best to use SI units and obtain a pressure in SI units (Pa), and then convert the pressure to the required units (Torr).

Solve

Expression (6.1) defines pressure as force divided by area.

$$P = \frac{F}{A}$$

The force exerted by the cylinder is its weight.

$$F = W = m \times g$$

The mass is 1.000 kg, and g (the acceleration due to gravity) is $9.81\ m\,s^{-2}$. The product of these two terms is the force in newtons.

$$F = m \times g = 1.000\ kg \times 9.81\ m\,s^{-2} = 9.81\ N$$

The force is exerted on the area of contact between the cylinder and the underlying surface. This circular area is calculated by using the radius of the cylinder—one-half the 4.10 cm diameter, expressed in meters.

$$A = \pi r^2 = 3.1416 \times \left(2.05\ cm \times \frac{1\ m}{100\ cm}\right)^2 = 1.32 \times 10^{-3}\ m^2$$

$Nm^{-2} = Pa$

The force divided by the area (in square meters) gives the pressure in pascals.

$$P = \frac{F}{A} = \frac{9.81\ N}{1.32 \times 10^{-3}\ m^2} = 7.43 \times 10^3\ Pa$$

The relationship between the units Torr and pascal (Table 6.1) is used for the final conversion.

$$P = 7.43 \times 10^3\ Pa \times \frac{760\ Torr}{101{,}325\ Pa} = 55.7\ Torr$$

Convert Pa to Torr



Class Practice 1: Comparing Liquid Pressures

Petrucci Example 6-1

What is the height of a column of water that exerts the same pressure as a column of mercury 76.0 cm (760 mm) high? Given that the d_{Hg} = 13.6 g/cm^3 and d_{H_2O} = 1.00 g/cm^3.

Class Practice 1: Comparing Liquid Pressures

Petrucci Example 6-1

What is the height of a column of water that exerts the same pressure as a column of mercury 76.0 cm (760 mm) high? Given that the d_{Hg} = 13.6 g/cm^3 and d_{H_2O} = 1.00 g/cm^3.

Solution: $P = h \times d \times g$

Analyze

Equation (6.2) shows that, for a given liquid pressure, the column height is inversely proportional to the liquid density. The lower the liquid density, the greater the height of the liquid column. Mercury is 13.6 times as dense as water (13.6 g/cm^3 versus 1.00 g/cm^3). If columns of water and mercury exert the same pressure, then the column of water is 13.6 times as high as the column of mercury.

Solve

Although we have already reasoned out the answer, we can arrive at the same conclusion by applying equation (6.2) twice, and then setting the two pressures equal to each other. Equation (6.2) can be used to describe the pressure of the mercury column of known height and the pressure of the water column of unknown height. Then we can set the two pressures equal to each other.

$$\text{pressure of Hg column} = g \times h_{Hg} \times d_{Hg} = g \times 76.0\,\text{cm} \times 13.6\,\text{g/cm}^3$$

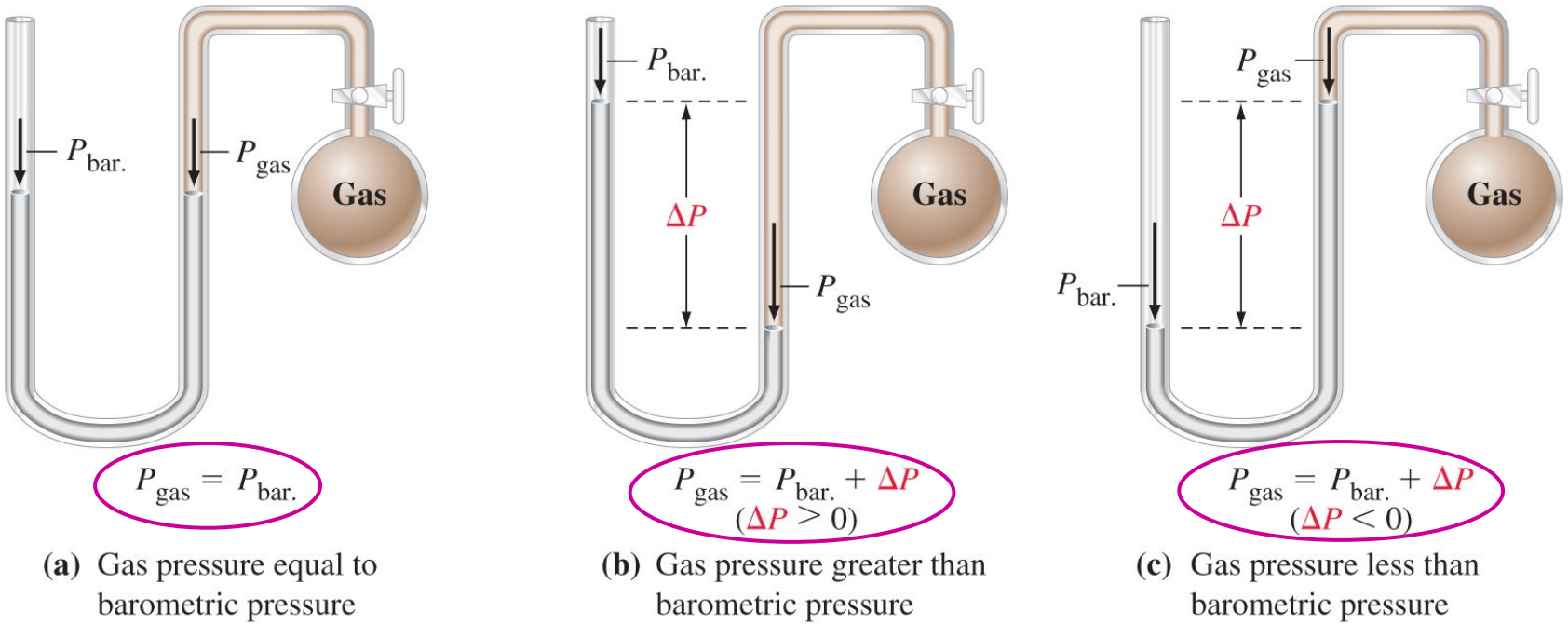
$$\text{pressure of } H_2O \text{ column} = g \times h_{H_2O} \times d_{H_2O} = g \times h_{H_2O} \times 1.00\,\text{g/cm}^3$$

$$g \times h_{H_2O} \times 1.00\,\text{g/cm}^3 = g \times 76.0\,\text{cm} \times 13.6\,\text{g/cm}^3$$

$$h_{H_2O} = 76.0\,\text{cm} \times \frac{13.6\,\text{g/cm}^3}{1.00\,\text{g/cm}^3} = 1.03 \times 10^3\,\text{cm} = 10.3\,\text{m}$$

Manometers

- A **manometer** is used to measure gas pressures.
- By using a manometer, the pressure of the gas to be measured can be compared with barometric pressure.



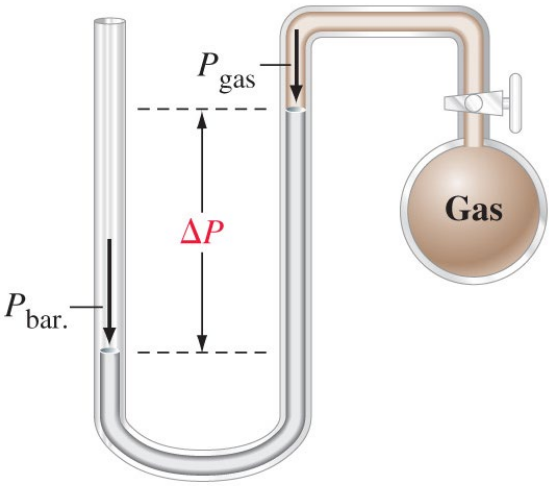
Measurement of gas pressure with an open-end manometer

The possible relationships between barometric pressure and a gas pressure under measurement are pictured here.

Class Practice 2

EXAMPLE 6-2 Using a Manometer to Measure Gas Pressure

When the manometer in Figure 6-5(c) is filled with liquid mercury ($d = 13.6\ \text{g/cm}^3$), the barometric pressure is 748.2 mmHg, and the difference in mercury levels is 8.6 mmHg. What is the gas pressure P_{gas}?



$$P_{gas} = P_{bar.} + \Delta P \quad (\Delta P < 0)$$

(c) Gas pressure less than barometric pressure

Class Practice 2 (Con't)

Analyze

We must first establish which is greater: the barometric pressure or the gas pressure. In Figure 6-5(c), the barometric pressure forces liquid mercury down the tube toward the gas sample. The barometric pressure is greater than the gas pressure. Thus, $\Delta P = P_{gas} - P_{bar.} < 0$.

Solve

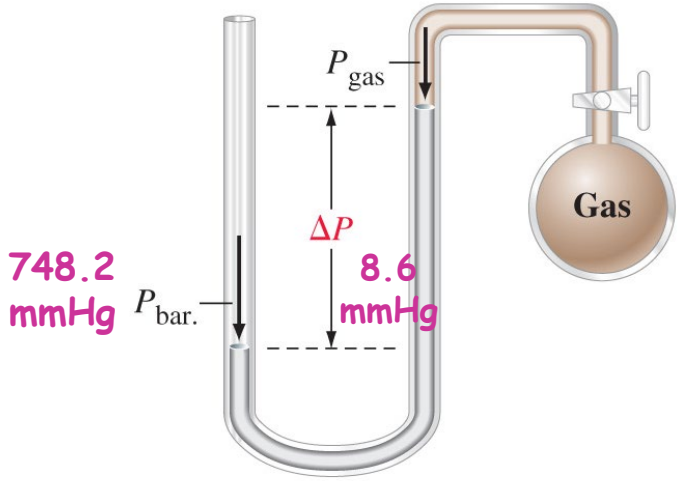
The gas pressure is less than the barometric pressure. Therefore, we subtract 8.6 mmHg from the barometric pressure to obtain the gas pressure.

$$P_{gas} = P_{bar.} + \Delta P = 748.2\ \text{mmHg} - 8.6\ \text{mmHg} = 739.6\ \text{mmHg}$$

Assess

Because all pressures are expressed in millimeters of mercury, the pressure difference (ΔP) is numerically equal to the difference in mercury levels. Thus, the density of mercury does not enter into the calculation.

$P = h \times d \times g$



$P_{gas} = P_{bar.} + \Delta P$
$(\Delta P < 0)$

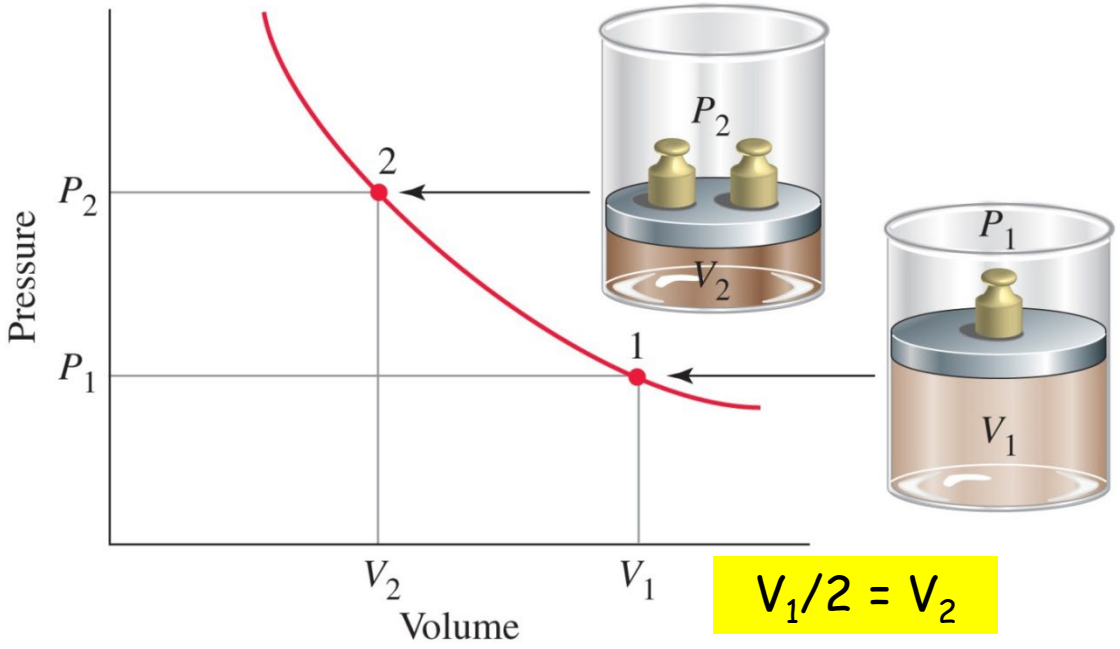
(c) Gas pressure less than barometric pressure

2. The Simple Gas Laws

(1) Boyle's Law (1662)

Scientific laws are statements, based on repeated experiments or observations, that describe or predict a range of natural phenomena

For a fixed amount of gas at a constant temperature, the gas volume is inversely proportional to the gas pressure.



$V_1/2 = V_2$

At constant n & T,

$$PV = \text{constant}$$

$$P \alpha \frac{1}{V}$$

Relationship between gas volume and pressure – Boyle's Law

A doubling of the pressure causes the volume to decrease to one-half its original value.

Class Practice 4

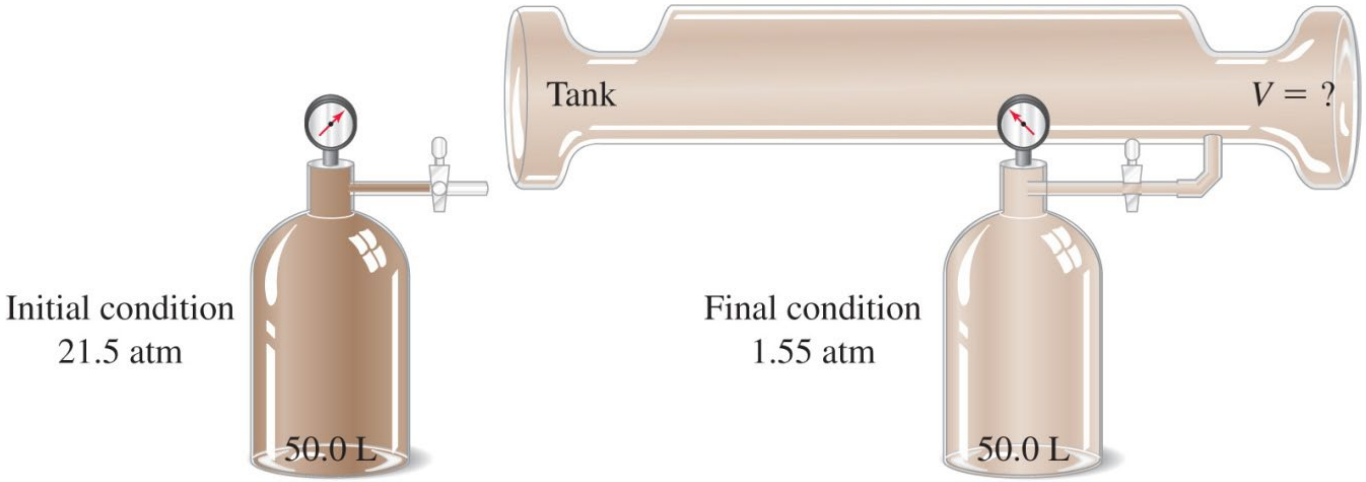
A 50.0 L cylinder contains nitrogen gas at a pressure of 21.5 atm. The cylinder is then connected to an evacuated tank of unknown volume. If the final pressure in the tank is 1.55 atm, then what is the volume of the tank?

(a) (21.5/1.55) x 50.0 L

(b) (1.55/21.5) x 50.0 L

(c) 21.5/(1.55 x 50.0) L

(d) 1.55/(21.5 x 50.0) L





Class Practice 4 (Con't)

Solution:

Use Boyle's Law

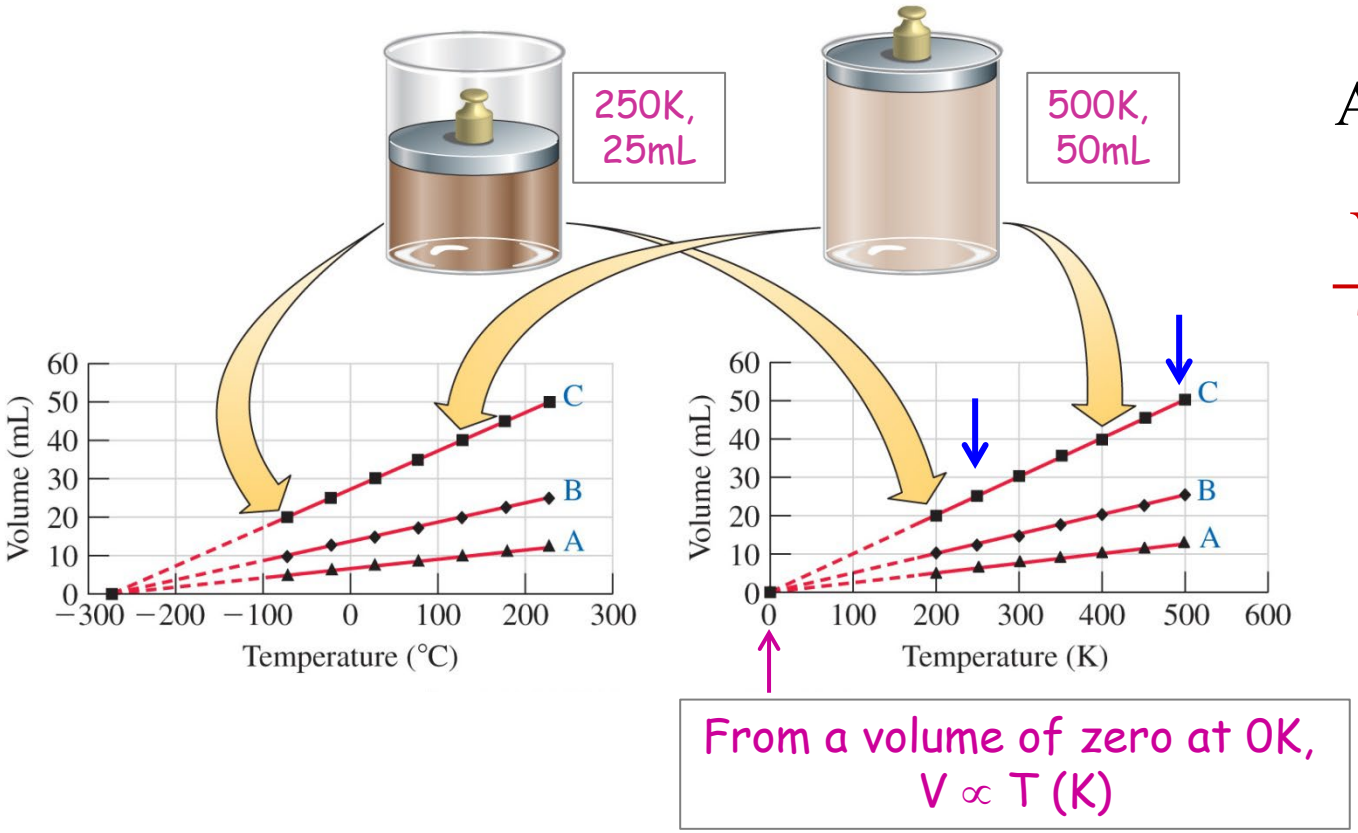
$$P_1V_1 = P_2V_2 \qquad V_2 = \frac{P_1V_1}{P_2}$$ **therefore (a) is the answer**

V_{total}= (21.5/1.55) x 50.0 L
= 693.5 L

V_{tank}= 693.5 L – 50.0L
= 643.5 L

(2) Charles's Law (1787)

The volume of a fixed amount of gas at a constant pressure is directly proportional to the Kelvin temperature.



At constant n & P,

$$\frac{V}{T} = \text{constant}$$

$$V \propto T$$

Gas volume as a function of temperature – Charles's Law

The volumes of three different gases (A, B, and C) are measured at 1 atm and 500 K. As the temperature is lowered, the volume decreases as predicted by Charles s law.

(3) Avogadro's Law (1811)

At a fixed temperature and pressure, the volume of a gas is directly proportional to the amount of gas.

At 0°C and 1 atm,

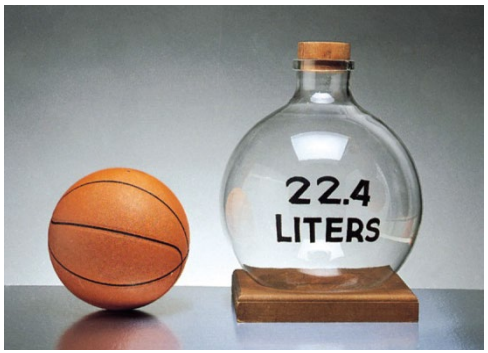
Helium 1 mole (4.0 g)	CH_4 1 mole (16.0 g)	CO_2 1 mole (44.0 g)
V = 22.4 dm³	**V = 22.4 dm³**	**V = 22.4 dm³**

Same molar volume (Theoretically)

At constant T & P,

$$\frac{V}{n} = \text{constant}$$

$$V \propto n$$



1 mol gas = 22.414 L (at 0 °C, 1 atm) = 22.711 L (STP)



Check Questions

(1) A sample of N_2 gas occupies 2.40 L at 20°C. If the gas is in a container that can contract or expand at constant pressure, at what temperature will the N_2 occupy 4.80 L?

A. 10°C
B. 40°C
C. 146°C
D. 313°C
E. 685°C

(2) 0.820 mole of hydrogen gas has a volume of 2.00 L at a certain temperature and pressure. What is the volume of 0.125 mol of this gas at the same temperature and pressure?

A. 0.0512 L
B. 0.250 L
C. 0.305 L
D. 4.01 L
E. 19.5 L



Check Questions

(1) A sample of N_2 gas occupies 2.40 L at 20°C. If the gas is in a container that can contract or expand at constant pressure, at what temperature will the N_2 occupy 4.80 L?

A. 10°C
B. 40°C
C. 146°C
D. 313°C
E. 685°C

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

T_2= 586 K = 313 °C (Answer: D)

(2) 0.820 mole of hydrogen gas has a volume of 2.00 L at a certain temperature and pressure. What is the volume of 0.125 mol of this gas at the same temperature and pressure?

A. 0.0512 L
B. 0.250 L
C. 0.305 L
D. 4.01 L
E. 19.5 L

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

V_2= 0.305 L(Answer: C)

3. Combining the Gas Laws:
The Ideal Gas Equation

- When the three laws combine, we obtain the ideal gas equation.

$$V \propto \frac{1}{P} \qquad V \propto T \qquad V \propto n$$

$$V \propto \frac{nT}{P}$$

$$PV = nRT$$

where R = Universal Gas Constant

For one mole of a gas at 273.15 K and 1 atm,

(101325 Pa) (2.24140 × 10^{-2} m^3) 8.3145 J/mol·K

$$R = \frac{(1\ atm)(22.4140L)}{(1\ mol)(273.15K)} = 0.082057\ \frac{atm \cdot L}{mol \cdot K}$$

Five Common Values of *R* (in various units)

$$R = \frac{PV}{nT}$$

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$

$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$

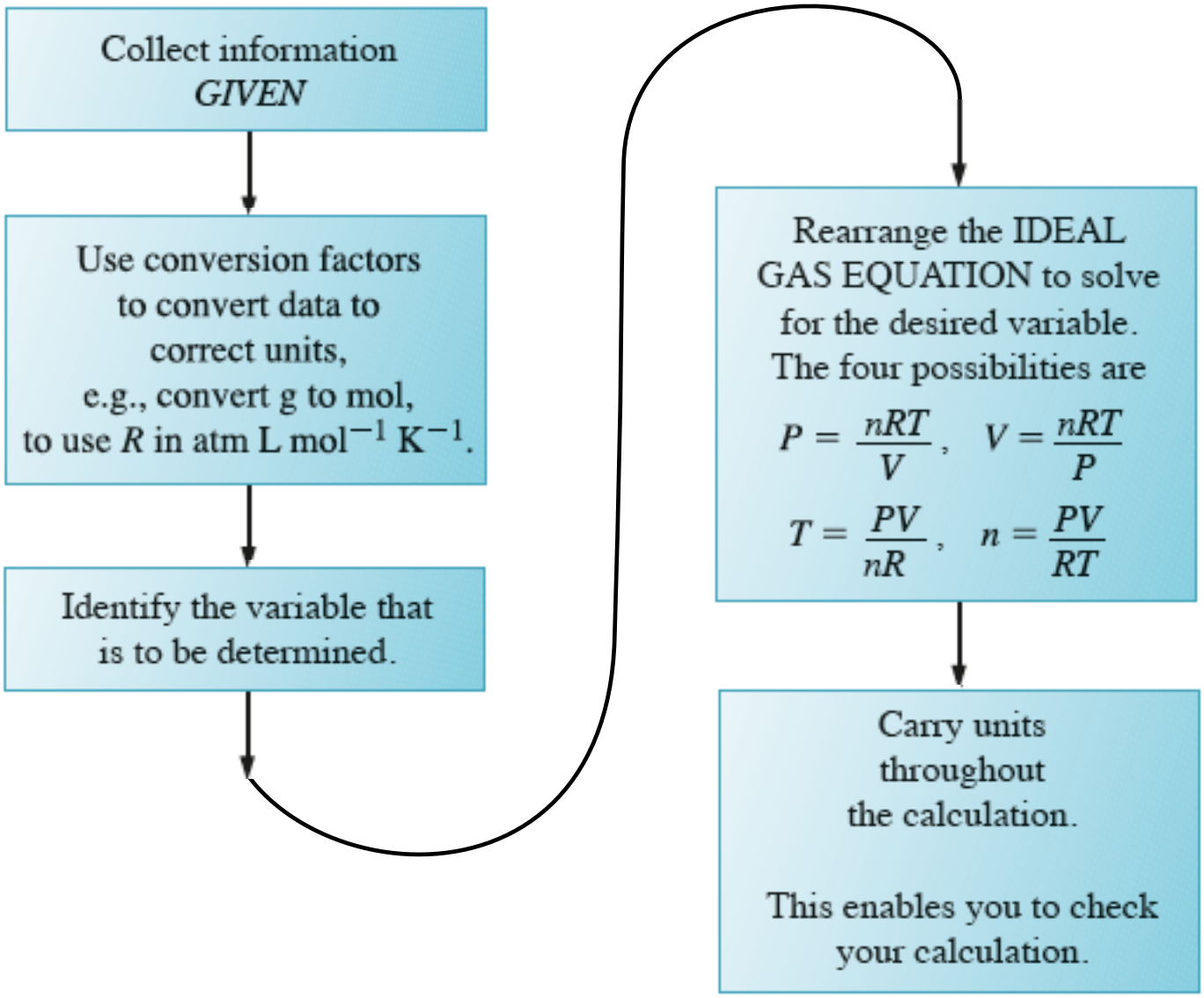
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$

$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$

$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

The value of R depends on which units are used to express pressure and volume.

Applying the ideal gas equation





Class Practice 5

EXAMPLE 6-4 Calculating a Gas Volume with the Ideal Gas Equation

What is the volume occupied by 13.7 g $Cl_2(g)$ at 45 °C and 98.4 kPa?

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

$$\frac{1\ \text{mol Cl}_2}{70.90\ \text{g Cl}_2}$$

Class Practice 5

EXAMPLE 6-4 Calculating a Gas Volume with the Ideal Gas Equation

What is the volume occupied by 13.7 g $Cl_2(g)$ at 45 °C and 98.4 kPa?

Solution: n✓ T ✓ P ✓ V?

Analyze

This is a relatively straightforward application of the ideal gas equation. We are given an amount of gas (in grams), a pressure (in kPa), and a temperature (in °C). Before using the ideal gas equation, we must express the amount in moles and the temperature in Kelvin. Include units throughout the calculation to ensure that the final result has acceptable units.

Solve

$P = 98.4\ \text{kPa}$

$V = ?$

$$n = 13.7\ \text{g Cl}_2 \times \frac{1\ \text{mol Cl}_2}{70.90\ \text{g Cl}_2} = 0.193\ \text{mol Cl}_2$$

$R = 8.314\ \text{kPa L mol}^{-1}\ \text{K}^{-1}$

$T = 45\ °\text{C} + 273 = 318\ \text{K}$

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

Divide both sides of the ideal gas equation by P to solve for V.

$$\frac{PV}{P} = \frac{nRT}{P} \quad \text{and} \quad V = \frac{nRT}{P}$$

$$V = \frac{nRT}{P} = \frac{0.193\ \text{mol} \times 8.314\ \text{kPa L mol}^{-1}\ \text{K}^{-1} \times 318\ \text{K}}{98.4\ \text{kPa}} = 5.19\ \text{L}$$



Class Practice 6

EXAMPLE 6-5 Calculating a Gas Pressure with the Ideal Gas Equation

What is the pressure, in kilopascals, exerted by 1.00×10^{20} molecules of N_2 in a 305 mL flask at 175 °C?

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

$$\frac{1\ \text{mol}\ N_2}{6.022 \times 10^{23}\ \text{molecules}\ N_2}$$

Class Practice 6

EXAMPLE 6-5 Calculating a Gas Pressure with the Ideal Gas Equation

What is the pressure, in kilopascals, exerted by 1.00×10^{20} molecules of N_2 in a 305 mL flask at 175 °C?

n✓ V ✓ T ✓ P (in kPa) ?

Solution:

Because we seek a pressure in kilopascals, let us use the form of the ideal gas equation having

$$R = 8.3145\ \text{kPa L mol}^{-1}\,\text{K}^{-1}$$

The first step is to convert from molecules to moles of a gas, n.

$$n = 1.00 \times 10^{20}\ \text{molecules } N_2 \times \frac{1\ \text{mol } N_2}{6.022 \times 10^{23}\ \text{molecules } N_2}$$

$$= 0.000166\ \text{mol } N_2$$

Convert from milliliters to liters and then to cubic meters (recall Figure 1-9).

$$V = 305\ \text{mL} \times \frac{1\ \text{L}}{1000\ \text{mL}} = 0.305\ \text{L}$$

Express gas temperature on the Kelvin scale.

$$T = (175 + 273)\ \text{K} = 448\ \text{K}$$

Rearrange the ideal gas equation to the form $P = nRT/V$, and substitute the above data.

$$P = \frac{nRT}{V} = \frac{0.000166\ \text{mol} \times 8.3145\ \text{kPa L mol}^{-1}\,\text{K}^{-1} \times 448\ \text{K}}{0.305\ \text{L}}$$

$$= 2.03\ \text{kPa}$$

The General Gas Equation

- The ideal gas equations must be applied ***twice*** in order to determine an initial condition and a final condition.

$$PV = nRT \qquad R = \frac{P_i V_i}{n_i T_i} = \frac{P_f V_f}{n_f T_f}$$

$$\frac{P_i V_i}{n_i T_i} = \frac{P_f V_f}{n_f T_f}$$ -- General Gas Equation

- This is often used in cases in which one or two of the gas properties are held constant, and the equation can be simplified by eliminating these constants.

E.g. For a constant amount of gas (fixed n),

$$\frac{P_i V_i}{T_i} = \frac{P_f V_f}{T_f}$$



Check Questions

(1) A sample of N_2 gas occupies 2.40 L at 20°C. If the gas is in a container that can contract or expand at constant pressure, at what temperature will the N_2 occupy 4.80 L?

A. 10°C
B. 40°C
C. 146°C
D. 313°C
E. 685°C

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

T_2= 586 K = 313 °C (Answer: D)

(2) 0.820 mole of hydrogen gas has a volume of 2.00 L at a certain temperature and pressure. What is the volume of 0.125 mol of this gas at the same temperature and pressure?

A. 0.0512 L
B. 0.250 L
C. 0.305 L
D. 4.01 L
E. 19.5 L

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

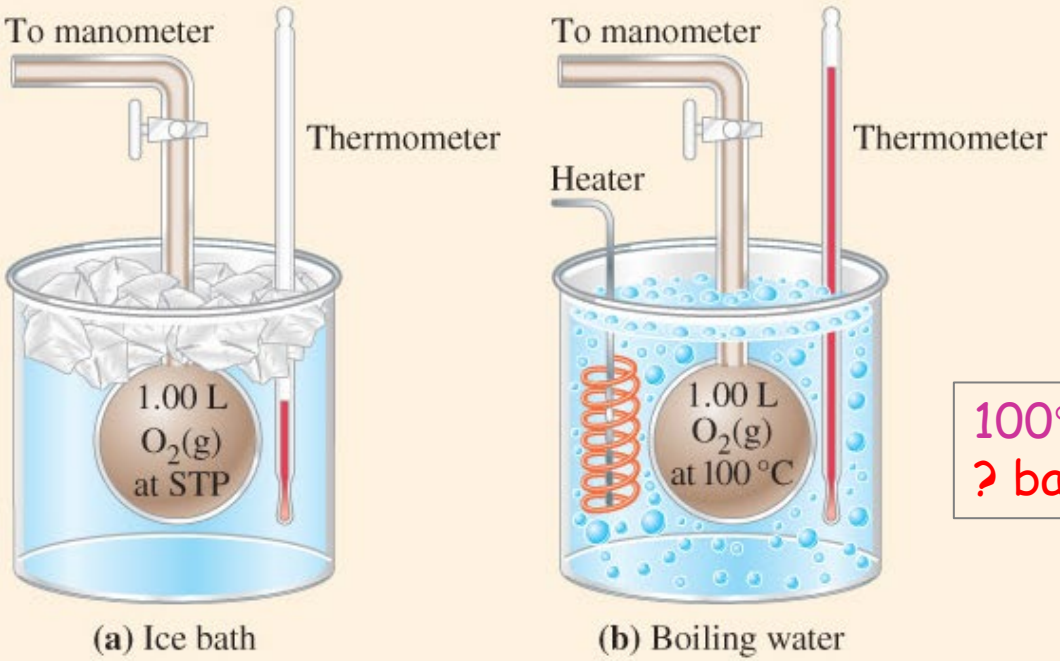
V_2= 0.305 L(Answer: C)

Class Practice 7

EXAMPLE 6-6 Applying the General Gas Equation

The situation pictured in Figure 6-10(a) is changed to that in Figure 6-10(b). What is the gas pressure in Figure 6-10(b)?

0°C
1 bar



(a) Ice bath (b) Boiling water

100°C
? bar

▲ FIGURE 6-10
Pressure of a gas as a function of temperature—Example 6-6 visualized
The amount of gas and volume are held constant.
(a) 1.00 L $O_2(g)$ at STP; **(b)** 1.00 L $O_2(g)$ at 100 °C.



Class Practice 7 (Con't)

Solution:

Analyze

Identify the quantities in the general gas equation that remain constant. Cancel out these quantities and solve the equation that remains.

Solve

In this case, the amount of O_2 is constant ($n_i = n_f$) and the volume is constant ($V_i = V_f$).

$$\frac{P_i \cancel{V_i}}{\cancel{n_i} T_i} = \frac{P_f \cancel{V_f}}{\cancel{n_f} T_f} \quad \text{and} \quad \frac{P_i}{T_i} = \frac{P_f}{T_f} \quad \text{and} \quad P_f = P_i \times \frac{T_f}{T_i}$$

Since $P_i = 1.00\,\text{bar}$, $T_i = 273\,\text{K}$, and $T_f = 373\,\text{K}$, then

$$P_f = 1.00\,\text{bar} \times \frac{373\,\text{K}}{273\,\text{K}} = 1.37\,\text{bar}$$

Assess

We can base our check on a qualitative, intuitive understanding of what happens when a gas is heated in a closed container. Its pressure increases (possibly to the extent that the container bursts). If, by error, we had used the ratio of temperatures 273 K/373 K, the final pressure would have been less than 1.00 bar—an impossible result.

Applications of the Ideal Gas Equation

1. Determination of molar masses
2. Determination of gas densities

Determination of molar masses

- If the volume of a gas (V) at a fixed T & P is known, we can obtain the amount of the gas in moles (n).

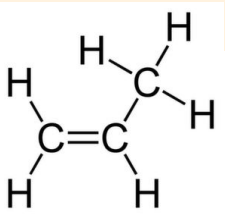
- Because $n = \frac{m}{M}$, if we know the mass (m) and no. of moles (n) of the gas, we can find out the molar mass (M).

$$PV = \frac{mRT}{M}$$

Class Practice 8

EXAMPLE 6-7 Determining a Molar Mass with the Ideal Gas Equation

Propylene is an important commercial chemical used in the synthesis of other organic chemicals and in production of plastics (polypropylene). A glass vessel weighs 40.1305 g when clean, dry, and evacuated; it weighs 138.2410 g when filled with water at 25.0 °C (density of water = 0.9970 g/mL) and 40.2959 g when filled with propylene gas at 740.3 mmHg and 24.0 °C. What is the molar mass of propylene?



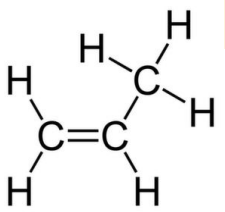
Atmosphere	atm	
Millimeter of mercury	mmHg	1 atm ≃ 760 mmHg
Torr	Torr	= 760 Torr
Pascal	Pa	= 101,325 Pa
Kilopascal	kPa	= 101.325 kPa
Bar	bar	= 1.01325 bar
Millibar	mbar	= 1013.25 mbar

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

Class Practice 8

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Solution:

Analyze

We are given a pressure (in mmHg), a temperature (in °C), and information that will enable us to determine the amount of gas (in grams) and the volume of the vessel. If we express these quantities in atmospheres, Kelvin, moles, and liters, respectively, then we can use equation (6.13), with $R = 0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$, to calculate the molar mass of the gas.

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

Class Practice 8 (Con't)

Solve

First determine the mass of water required to fill the vessel.

$$\text{mass of water to fill vessel} = 138.2410\ \text{g} - 40.1305\ \text{g} = 98.1105\ \text{g}$$

Use the density of water in a conversion factor to obtain the volume of water (and hence, the volume of the glass vessel).

$$\text{volume of water (volume of vessel)} = 98.1105\ \text{g}\ H_2O \times \frac{1\ \text{mL}\ H_2O}{0.9970\ \text{g}\ H_2O} = 98.41\ \text{mL} = 0.09841\ \text{L}$$

The mass of the gas is the difference between the weight of the vessel filled with propylene gas and the weight of the empty vessel.

$$\text{mass of gas} = 40.2959\ \text{g} - 40.1305\ \text{g} = 0.1654\ \text{g}$$

The values of temperature and pressure are given.

$$T = (24.0 + 273.15)\ \text{K} = 297.2\ \text{K}$$

$$P = 740.3\ \text{mmHg} \times \frac{1\ \text{atm}}{760\ \text{mmHg}} = 0.9741\ \text{atm}$$

Substitute data into the rearranged version of equation (6.13).

$$M = \frac{mRT}{PV} = \frac{0.1654\ \text{g} \times 0.082057\ \cancel{\text{atm}}\ \cancel{\text{L}}\ \text{mol}^{-1}\ \cancel{\text{K}^{-1}} \times 297.2\ \cancel{\text{K}}}{0.9741\ \cancel{\text{atm}} \times 0.09841\ \cancel{\text{L}}}$$

$$= 42.08\ \text{g mol}^{-1}$$

Determination of gas densities

- The density of a gas at STP can be calculated by dividing its molar mass by the molar volume (22.7 L/mol).

E.g. For O_2 at STP, the density is 32.0 g/22.7 L = 1.41 g/L

$$d = \frac{m}{V} = \frac{n \times M}{V} = \frac{n}{V} \times M$$

KEEP IN MIND

that gas densities are typically much smaller than those of liquids and solids. Gas densities are usually expressed in grams per *liter* rather than grams per *milliliter.*

- Replace n/V by its **equivalent**, P/RT,

$$d = \frac{m}{V} = \frac{MP}{RT}$$

d ∝ M (for gas)
d ∝ 1/T (for gas)





1 mol gas = 22.414 L (at 0 °C, 1 atm) and 22.711 L (at 0 °C, 1 bar, STP)



Class Practice 9

EXAMPLE 6-8 Using the Ideal Gas Equation to Calculate a Gas Density

What is the density of oxygen gas (O_2) at 298 K and 0.987 bar?

Class Practice 9

EXAMPLE 6-8 Using the Ideal Gas Equation to Calculate a Gas Density

What is the density of oxygen gas (O_2) at 298 K and 0.987 bar?

Solution:

Analyze

The gas is identified, and therefore the molar mass can be calculated. We are given a temperature in Kelvin and a pressure in bars, so we can use equation (6.14) directly with $R = 0.08314\ \text{bar L mol}^{-1}\,\text{K}^{-1}$.

Solve

The molar mass of O_2 is $32.0\ \text{g mol}^{-1}$. Now, use equation (6.14).

$$d = \frac{m}{V} = \frac{MP}{RT} = \frac{32.00\ \text{g mol}^{-1} \times 0.987\ \text{bar}}{0.08314\ \text{bar L mol}^{-1}\,\text{K}^{-1} \times 298\ \text{K}} = 1.27\ \text{g/L}$$

4. Gases in Chemical Reactions

Law of Combining Volumes

$$2NO\ (g) + O_2\ (g) \rightarrow 2NO_2\ (g)$$

$$2\ mol + 1\ mol \rightarrow 2\ mol$$

At constant T & P, $2\ V\ L + 1\ V\ L \rightarrow 2\ V\ L$

$$2\ L\ NO\ (g) + 1\ L\ O_2\ (g) \rightarrow 2\ L\ NO_2\ (g)$$

- The volume ratio of gases consumed and produced in a chemical reaction is the same as the mole (small-whole numbers) ratio (at constant T & P) - - *Gay-Lussac's Law of Combining Volumes*

Recall: Avogadro's Law

At constant T & P, $V \propto n$

Gay-Lussac's Law (1808)
Avogadro's Law (1811)

E.g.) Applying the Law of Combining Volumes

Zinc blende, ZnS, is the most important zinc ore. Roasting (strong heating) of ZnS in oxygen is the first step in the commercial production of zinc.

$$2\,ZnS(s) + 3\,O_2(g) \xrightarrow{\Delta} 2\,ZnO(s) + 2\,SO_2(g)$$

What volume of $SO_2(g)$ can be obtained from 1.00 L $O_2(g)$ and excess ZnS(s)? Both gases are measured at 25 °C and 98.0 kPa.

- The volume ratio of gases consumed and produced in a chemical reaction is the same as the mole (small-whole numbers) ratio (at constant T & P) - - *Gay-Lussac's Law of Combining Volumes*

E.g.) Applying the Law of Combining Volumes

Petrucci Example 6-10

Zinc blende, ZnS, is the most important zinc ore. Roasting (strong heating) of ZnS in oxygen is the first step in the commercial production of zinc.

1L ?L

$$2\,ZnS(s) + 3\,O_2(g) \xrightarrow{\Delta} 2\,ZnO(s) + 2\,SO_2(g)$$

What volume of $SO_2(g)$ can be obtained from 1.00 L $O_2(g)$ and excess ZnS(s)? Both gases are measured at 25 °C and 98.0 kPa.

Analyze

The reactant and product being compared are both gases, and both are at the same temperature and pressure. Therefore, we can use the law of combining volumes and treat the coefficients in the balanced chemical equation as if they had units of liters.

$3O_2\,(g) : 2SO_2\,(g)$

Solve

The stoichiometric factor (shown below in blue) converts from L $O_2(g)$ to L $SO_2(g)$.

$$?\,L\,SO_2(g) = 1.00\,L\,O_2(g) \times \frac{2\,L\,SO_2(g)}{3\,L\,O_2(g)} = 0.667\,L\,SO_2(g)$$

Assess

Some students would solve this problem by using the following sequence of conversions: L O_2 ⟶ mol O_2 ⟶ mol SO_2 ⟶ L SO_2. This approach is acceptable but not as simple as the approach we used.

Gay-Lussac's Law + Avogadro's Law

E.g. $2NaN_3\ (s) \xrightarrow{\Delta} 2Na\ (l) + 3N_2\ (g)$

- For reactions involving gases, we can

  (1) use stoichiometric factors to relate gas quantities to quantities of other reactants or products,

  (2) use the ideal gas equation to relate the amount of a gas to volume, temperature and pressure.



Class Practice 10

EXAMPLE 6-9 Using the Ideal Gas Equation in Reaction Stoichiometry Calculations

What volume of N_2, measured at 98.0 kPa and 26 °C, is produced when 75.0 g NaN_3 is decomposed?

$$2\,NaN_3(s) \xrightarrow{\Delta} 2\,Na(l) + 3\,N_2(g)$$

65.1 g/mol (NaN_3)

$0.082057\ atm\ L\ mol^{-1}\ K^{-1}$
$0.083145\ bar\ L\ K^{-1}\ mol^{-1}$
$8.3145\ kPa\ L\ K^{-1}\ mol^{-1}$
$8.3145\ Pa\ m^3\ mol^{-1}\ K^{-1}$
$8.3145\ J\ mol^{-1}\ K^{-1}$

Class Practice 10

EXAMPLE 6-9 Using the Ideal Gas Equation in Reaction Stoichiometry Calculations

What volume of N_2, measured at 98.0 kPa and 26 °C, is produced when 75.0 g NaN_3 is decomposed?

$$2\,NaN_3(s) \xrightarrow{\Delta} 2\,Na(l) + 3\,N_2(g)$$

n (indirectly)	P ✓	T ✓	V ?

Solution:

Analyze

The following conversions are required.

$$g\,NaN_3 \longrightarrow mol\,NaN_3 \longrightarrow mol\,N_2 \longrightarrow L\,N_2$$

The molar mass of NaN_3 is used for the first conversion. The second conversion makes use of a stoichiometric factor constructed from the coefficients of the chemical equation. The ideal gas equation is used to complete the final conversion.

Class Practice 10 (Con't)

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

$$?\ \text{mol N}_2 = 75.0\ \text{g NaN}_3 \times \frac{1\ \text{mol NaN}_3}{65.01\ \text{g NaN}_3} \times \frac{3\ \text{mol N}_2}{2\ \text{mol NaN}_3} = 1.73\ \text{mol N}_2$$

$P = 98.0\ \text{kPa}$

$V = ?$

$n = 1.73\ \text{mol}$

$R = 8.314\ \text{kPa L mol}^{-1}\ \text{K}^{-1}$

$T = (26 + 273)\ \text{K} = 299\ \text{K}$

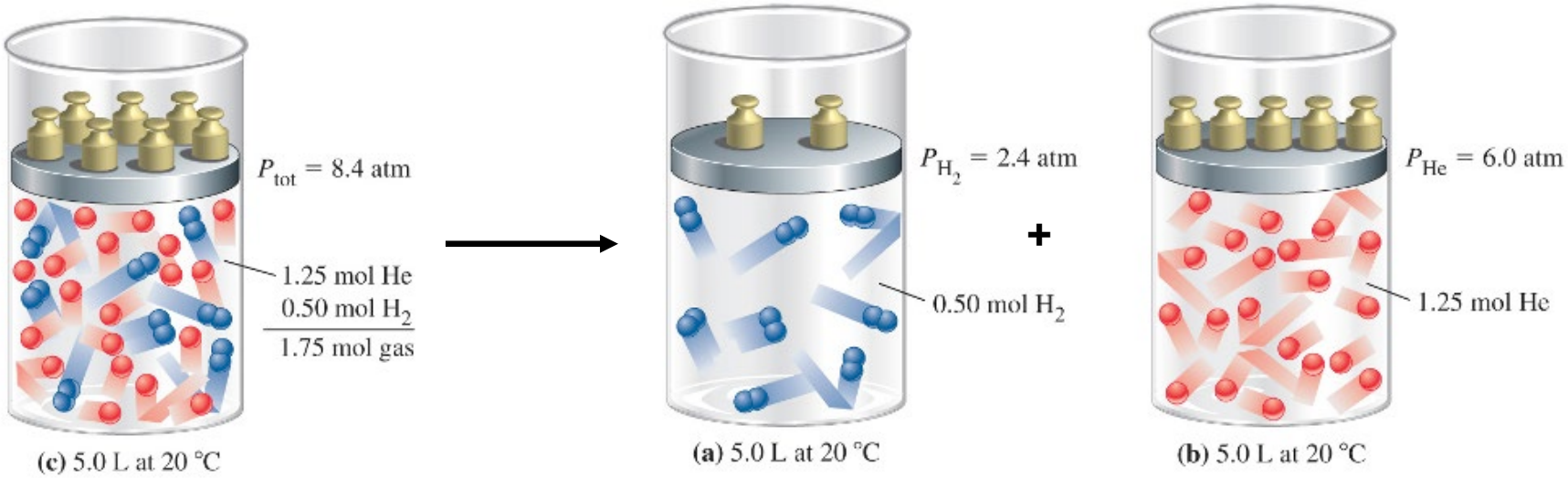
$$V = \frac{nRT}{P} = \frac{1.73\ \text{mol} \times 8.314\ \text{kPa L mol}^{-1}\ \text{K}^{-1} \times 299\ \text{K}}{98.0\ \text{kPa}} = 43.9\ \text{L}$$

5. Mixtures of Gases

All gases are the same, ideal gas.

Dalton's Law of Partial Pressures

The pressure of each gas is proportional to the number of moles of gas. The total pressure is the sum of the partial pressures of the individual gases.



Dalton's law of partial pressures illustrated

$$P_A = \frac{n_A}{n_{tot}} P_{tot}$$

$$P_{tot} = P_A + P_B + \ldots$$

$$PV = nRT$$

5. Mixtures of Gases

- Gas laws apply to *mixtures* of (nonreactive) gases as well as to individual gases. PV = nRT
- Simplest approach is to use the total number of moles of the gaseous mixture, n_{total}.

E.g.) Applying Ideal Gas Eqn to a Gas Mixture

Petrucci Example 6-11

What is the pressure, in bar, exerted by a mixture of 1.0 g H_2 and 5.00 g He when the mixture is confined to a volume of 5.0 L at 20 °C?

n_{total} ✓ V ✓ T ✓ P (in bar) ?

Analyze

For fixed T and V, the total pressure of a mixture of gases is determined by the total number of moles of gas: $P_{tot} = n_{tot}RT/V$.

Solve

$$n_{tot} = \left(1.0\ \text{g H}_2 \times \frac{1\ \text{mol H}_2}{2.02\ \text{g H}_2}\right) + \left(5.00\ \text{g He} \times \frac{1\ \text{mol He}}{4.003\ \text{g He}}\right)$$

$$= 0.50\ \text{mol H}_2 + 1.25\ \text{mol He} = 1.75\ \text{mol gas}$$

P = nRT/V

$$P_{tot} = \frac{1.75\ \text{mol} \times 0.0831\ \text{bar L mol}^{-1}\ \text{K}^{-1} \times (20 + 273)\ \text{K}}{5.0\ \text{L}} = 8.5\ \text{bar}$$

$$\frac{P_i V_i}{n_i T_i} = \frac{P_f V_f}{n_f T_f}$$

Assess

It is also possible to solve this problem by starting from equation (6.12). Because 1 mol of ideal gas occupies 22.7 L at 273 K and 1 bar, the pressure exerted by 1.75 mol of gas in a 5.0 L vessel at 293 K is (1.75 mol/1.00 mol) × (293 K/273 K) × (22.7 L/5.0 L) × 1.0 bar = 8.5 bar.

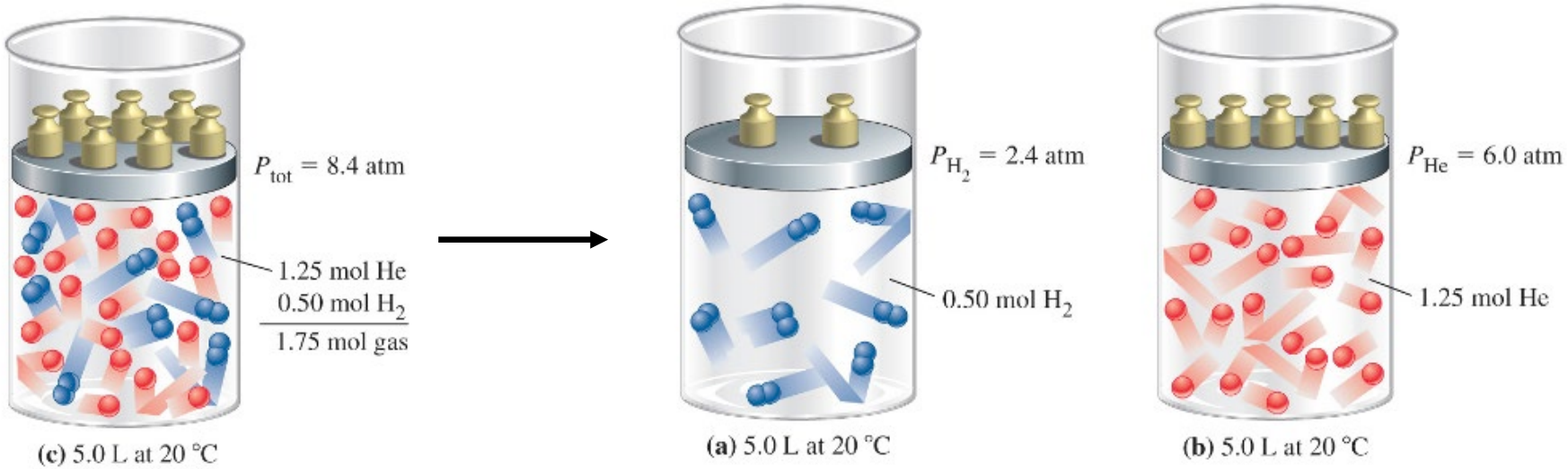
Partial Pressures

- For a mixture of gases, A, B, and so on,

$$P_{tot} = P_A + P_B + \ldots$$

$$\frac{P_A}{P_{tot}} = \frac{n_A RT/V_{tot}}{n_{tot} RT/V_{tot}} = \frac{n_A}{n_{tot}}$$

The total pressure is the sum of partial pressures.



Partial Pressures

The mole fraction of A, χ_A

$$\frac{n_A}{n_{tot}} = \frac{P_A}{P_{tot}} = \frac{V_A}{V_{tot}} = x_A$$

$$PV = nRT$$

- The mole fraction of a component in a mixture is the fraction of all the molecules in the mixture contributed by that component.
- The sum of all the mole fractions in a mixture is one.

Partial volume

- In a gaseous mixture of n_A moles of A, n_B moles of B, and so on, the volume each gas would individually occupy at a pressure equal to P_{tot} is

$$V_A = \frac{n_A}{n_{tot}} V_{tot} \qquad \frac{V_A}{V_{tot}} = \frac{n_A RT/P_{tot}}{n_{tot} RT/P_{tot}} = \frac{n_A}{n_{tot}}$$

- The total volume of the gaseous mixture is

$$V_{tot} = V_A + V_B + \cdots$$

The total volume is the sum of individual volumes.

Partial Pressures and volume

- For a mixture of gases, A, B, and so on,

$$P_{tot} = P_A + P_B + \ldots$$

The total pressure is the sum of partial pressures.

- In a gaseous mixture of n_A moles of A, n_B moles of B, and so on, the volume each gas would individually occupy at a pressure equal to P_{tot} is $V_A = n_A RT/P_{tot}$

The total volume is the sum of individual volumes.

- The total volume of the gaseous mixture is $V_{tot} = V_A + V_B + \ldots$

- We can derive a particularly useful expression from the following ratios,

$$\frac{P_A}{P_{tot}} = \frac{n_A RT/V_{tot}}{n_{tot} RT/V_{tot}} = \frac{n_A}{n_{tot}} \qquad \frac{V_A}{V_{tot}} = \frac{n_A RT/P_{tot}}{n_{tot} RT/P_{tot}} = \frac{n_A}{n_{tot}}$$

The mole fraction of A, χ_A

$$\frac{n_A}{n_{tot}} = \frac{P_A}{P_{tot}} = \frac{V_A}{V_{tot}} = x_A$$



Class Practice 11

EXAMPLE 6-12 Calculating the Partial Pressures in a Gaseous Mixture

What are the partial pressures of H_2 and He in the gaseous mixture described in Example 6-11?

0.50 mol H_2 and 1.25 mol He
5.0 L and 20 °C

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

Class Practice 11

EXAMPLE 6-12 Calculating the Partial Pressures in a Gaseous Mixture

What are the partial pressures of H_2 and He in the gaseous mixture described in Example 6-11?

0.50 mol H_2 and 1.25 mol He
5.0 L and 20 °C

Solution:

Analyze

The ideal gas equation can be applied to each gas individually to obtain the partial pressure of each gas.

Solve

One approach involves a direct application of Dalton's law in which we calculate the pressure that each gas would exert if it were alone in the container.

$$P_{H_2} = \frac{n_{H_2} \times RT}{V} = \frac{0.50\ \text{mol} \times 0.0831\ \text{bar L mol}^{-1}\ \text{K}^{-1} \times 293\ \text{K}}{5.0\ \text{L}} = 2.4\ \text{bar}$$

$$P_{He} = \frac{n_{He} \times RT}{V} = \frac{1.25\ \text{mol} \times 0.0831\ \text{bar L mol}^{-1}\ \text{K}^{-1} \times 293\ \text{K}}{5.0\ \text{L}} = 6.1\ \text{bar}$$

Or

Expression (6.17) gives us a simpler way to answer the question because we already know the number of moles of each gas and the total pressure from Example 6-11 ($P_{tot} = 8.4$ atm).

$$\frac{n_A}{n_{tot}} = \frac{P_A}{P_{tot}} = \frac{V_A}{V_{tot}} = x_A$$

$$P_{H_2} = \frac{n_{H_2}}{n_{tot}} \times P_{tot} = \frac{0.50}{1.75} \times 8.5\ \text{bar} = 2.4\ \text{bar}$$

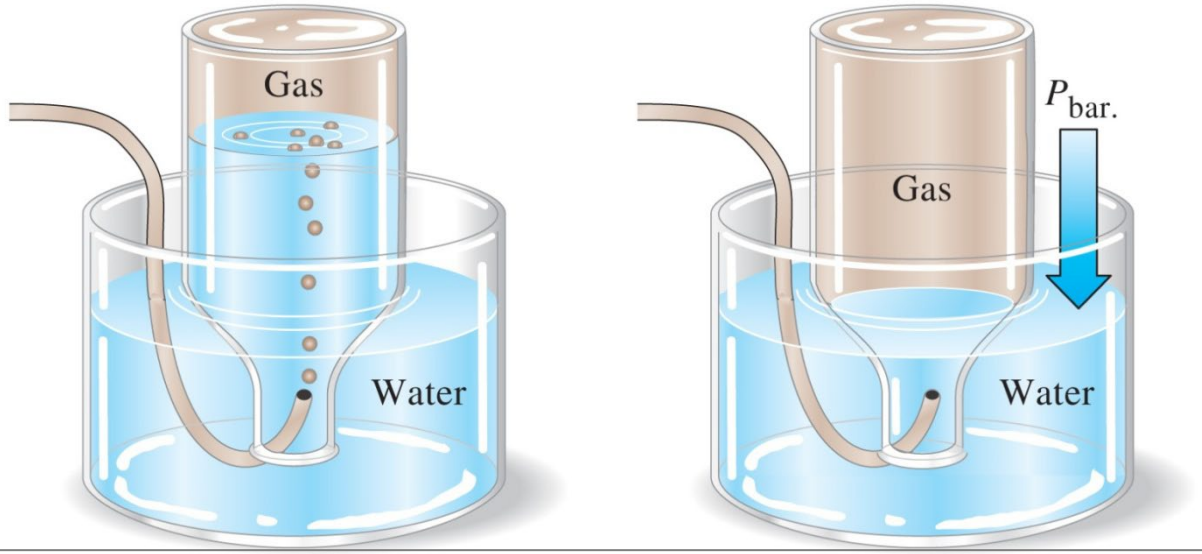
$$P_{He} = \frac{n_{He}}{n_{tot}} \times P_{tot} = \frac{1.25}{1.75} \times 8.5\ \text{bar} = 6.1\ \text{bar}$$

Collecting a Gas over Water

$$P_{tot} = P_{bar} = P_{gas} + P_{H_2O}$$

A pneumatic trough

For gases that are **insoluble** in and do **not** react with the liquid being displaced.



A gas collected this way is "wet" (∵ contains water vapor).

- The bottle is filled with water and its open end is held below the water level in the container.
- Gas from a gas-generating apparatus is directed into the bottle.

∵ P_{gas} ↑

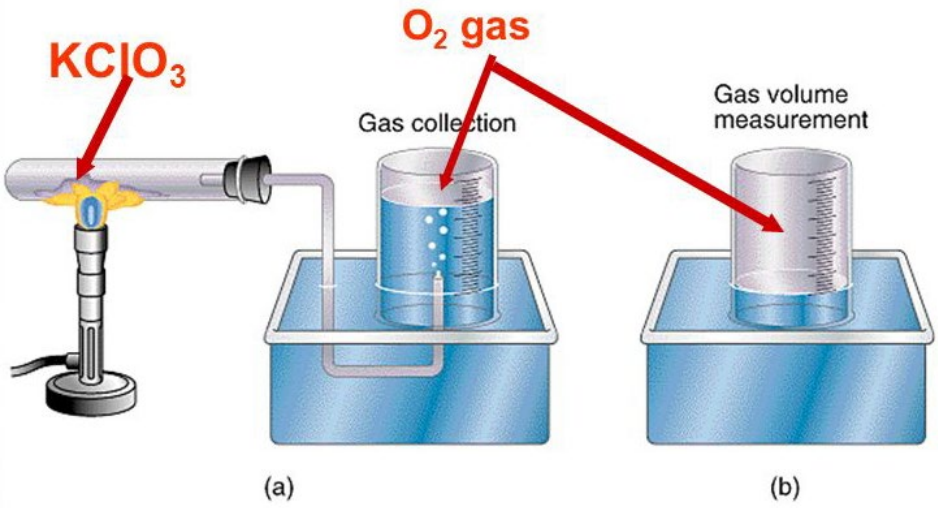
- As gas accumulates in the bottle, water is displaced from the bottle into the container.
- To make the total gas pressure in the bottle equal to barometric pressure, the position of the bottle must be adjusted so that the water levels inside and outside the bottle are the same.

$$P_{tot} = P_{bar} = P_{gas} + P_{H_2O}$$

Pressure of the atmosphere

The pressure of water vapour depends on the temp.

Collecting a Gas over Water



Vapour Pressure of Water at Various Temperatures

	Vapor Pressure	
t, °C	kPa	mmHg
15.0	1.706	12.79
17.0	1.938	14.54
19.0	2.198	16.49
21.0	2.448	18.66
23.0	2.811	21.08
25.0	3.170	23.78
30.0	4.247	31.86
50.0	12.35	92.65

Class Practice 12

EXAMPLE 6-13 Collecting a Gas over a Liquid (Water)

In the following reaction, 81.2 mL of $O_2(g)$ is collected over water at 23 °C and barometric pressure 751 mmHg. What mass of $Ag_2O(s)$ decomposed? (The vapor pressure of water at 23 °C is 21.1 mmHg.)

$$2\ Ag_2O(s) \longrightarrow 4\ Ag(s) + O_2(g)$$

$$\frac{231.7\ g\ Ag_2O}{1\ mol\ Ag_2O}$$

$$0.082057\ atm\ L\ mol^{-1}\ K^{-1}$$
$$0.083145\ bar\ L\ K^{-1}\ mol^{-1}$$
$$8.3145\ kPa\ L\ K^{-1}\ mol^{-1}$$
$$8.3145\ Pa\ m^3\ mol^{-1}\ K^{-1}$$
$$8.3145\ J\ mol^{-1}\ K^{-1}$$

Class Practice 12

EXAMPLE 6-13 Collecting a Gas over a Liquid (Water)

In the following reaction, 81.2 mL of $O_2(g)$ is collected over water at 23 °C and barometric pressure 751 mmHg. What mass of $Ag_2O(s)$ decomposed? (The vapor pressure of water at 23 °C is 21.1 mmHg.)

$$2\ Ag_2O(s) \longrightarrow 4\ Ag(s) + O_2(g)$$

Solution:

Mass of Ag_2O = ? ← **n_{O_2} = ?**

Analyze

The key concept is that the gas collected is wet, that is, a *mixture* of $O_2(g)$ and water vapor. Use $P_{bar.} = P_{O_2} + P_{H_2O}$ to calculate P_{O_2}, and then use the ideal gas equation to calculate the number of moles of O_2. The following conversions are used to complete the calculation: mol O_2 ⟶ mol Ag_2O ⟶ g Ag_2O.

Solve

$$P_{O_2} = P_{bar.} - P_{H_2O} = 751\ mmHg - 21.1\ mmHg = 730\ mmHg$$

$$P_{O_2} = 730\ mmHg \times \frac{1\ atm}{760\ mmHg} = 0.961\ atm$$

$$V = 81.2\ mL = 0.0812\ L$$

$$n = ?$$

$$R = 0.08206\ atm\ L\ mol^{-1}\ K^{-1}$$

$$T = (23 + 273)\ K = 296\ K$$

$$n = \frac{PV}{RT} = \frac{0.961\ atm \times 0.0812\ L}{0.08206\ atm\ L\ mol^{-1}\ K^{-1} \times 296\ K} = 0.00321\ mol$$

From the chemical equation we obtain a factor to convert from moles of O_2 to moles of Ag_2O. The molar mass of Ag_2O provides the final factor.

$$?\ g\ Ag_2O = 0.00321\ mol\ O_2 \times \frac{2\ mol\ Ag_2O}{1\ mol\ O_2} \times \frac{231.7\ g\ Ag_2O}{1\ mol\ Ag_2O} = 1.49\ g\ Ag_2O$$

Class Practice 12 (Con't)

Assess

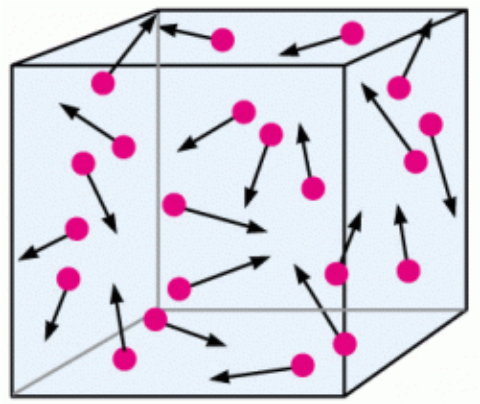
The determination of the number of moles of O_2 in the sample is the key calculation. We can quickly estimate the number of moles of O_2 in the sample by using the fact that for typical conditions ($T \approx 298\,K, P \approx 760\,mmHg$), the molar volume of an ideal gas is about 24 L. The number of moles of gas (mostly O_2) in the sample is approximately $0.08\,L/24\,L\,mol^{-1} \approx 0.003\,mol$. This estimate is quite close to the value calculated above.

6. Kinetic-Molecular Theory of Gases

- The simple gas laws and the ideal gas equation are used to predict gas behaviour. They are natural laws.

- To explain the gas laws, we need a ***theory***.

- **Kinetic-Molecular Theory of Gases** was developed in mid-19th century.



Macroscopic properties of a gas (P, V, T)



Microscopic properties of the gas molecules (x, u)

Kinetic-Molecular Theory of Gases

- A gas is composed of a very large number of particles in constant, random, straight-line motion.
- Particles are point masses and separated by great distances.
- Collisions are rapid and elastic.

> An **elastic collision** is one that there is no loss of kinetic energy in the collision.

- No force between particles.
- Individual particles may gain or lose energy as a result of collisions. But total energy remains constant.



Visualizing Molecular Motion

The validity of this model can be checked by comparing predictions based on the model with experimental facts. It was proven that **the predictions based on this model are consistent with several observed macroscopic properties**.

Derivation of Boyle's Law

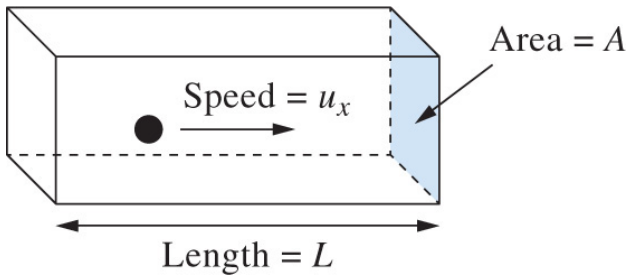
- Kinetic-Molecular Theory of Gases provides a satisfactory explanation of Boyle's law.

PV = a

The value of 'a' depends on N and T of the sample.

- Because **P is F/A**, the key to deriving the above equation is in assessing the forces associated with molecules hitting the walls of the container.

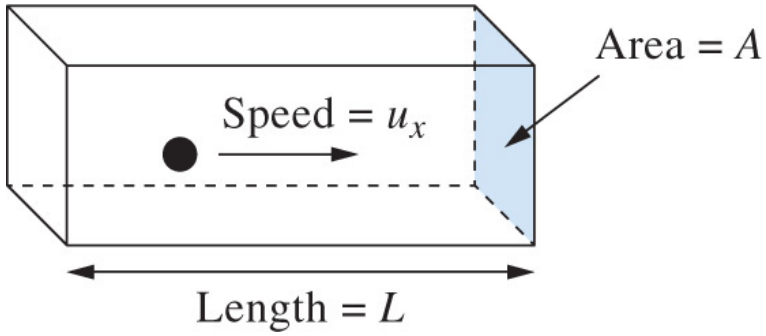
- Consider a molecule travelling along the **x** direction toward a wall perpendicular to it path. The speed of the molecule is denoted by $\mathbf{u_x}$. **The force exerted on the wall** by the molecule depend on the following:



↑ freq ⇒ ↑ total force

1. The collision frequency – the no. of collisions per unit time.

2. The momentum transfer, or impulse per collision (momentum is transferred as the molecule reverses direction upon hitting the container wall). The momentum transfer to the wall is $+2mu_x$.



A model for calculating the pressure exerted by a single molecule

A particle moving with speed u_x toward the end wall of a cubical box has initial momentum mu_x. After deflecting off the wall, it has final momentum $-mu_x$. The change in momentum is $-mu_x - mu_x = \mathbf{-2mu_x}$. The time between collisions is $\mathbf{2L/u_x}$.

And the collision frequency is $u_x/2L$.

- The force exerted by the molecule on the wall is the product of the momentum transfer and the collision frequency:

$(+2mu_x)(u_x/2L) = mu_x^2/L$ Force = Newton (N) = kg m/s^2

$$\therefore \text{Pressure} = \frac{\text{force}}{\text{area}} = \frac{mu_x{}^2/L}{A} = \frac{mu_x{}^2}{AL} = \frac{mu_x{}^2}{V}$$ (for one molecule)

- For a sample of gas containing *N* molecules each traveling in the x direction but with different speeds. We multiply the previous expression by *N* and replace $u_x{}^2$ with $\overline{u_x{}^2}$, the expression for pressure becomes:

mean square speed in x

$$P = \frac{N}{V} m\overline{u_x{}^2}$$

$$P = \frac{N}{V} m\overline{u_x{}^2}$$

$\because$ $\overline{u_x{}^2} = \overline{u_y{}^2} = \overline{u_z{}^2} = \frac{1}{3}\overline{u^2}$, where $\overline{u^2} = \overline{u_x{}^2} + \overline{u_y{}^2} + \overline{u_z{}^2}$

- $\overline{u^2}$ is called the mean-square speed. The average value of u^2 taking into account all the molecules, not just those moving in the x direction.

- Substitute $\frac{1}{3}\overline{u^2}$ for $\overline{u_x{}^2}$ into the above expression for P, and convert the expression into an equation because the factor of 1/3 is the proportionality constant, Thus, we get,

$$P = \frac{1}{3}\frac{N}{V} m\overline{u^2}$$

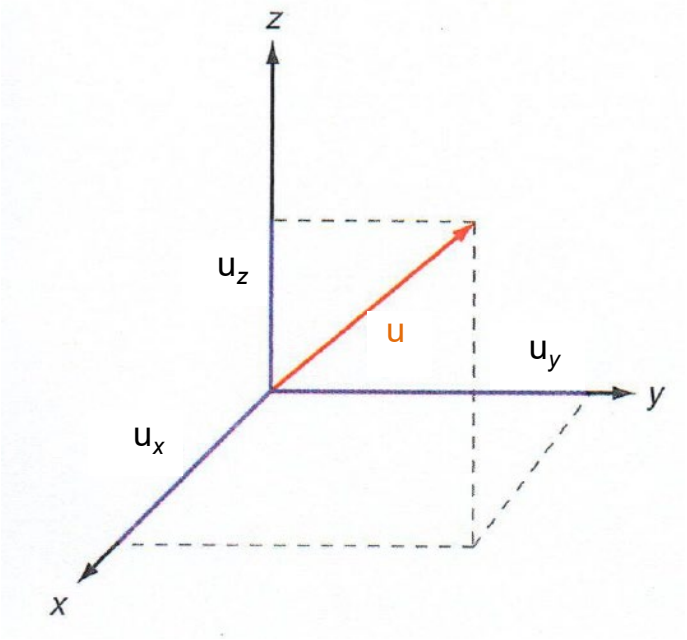
mean-square speed

-- eqn 3

-- this is the basic equation of the kinetic-molecular theory of gases.

Supplementary Information on Cartesian Components of Speed

- To describe pressure using kinetic theory, a molecule is envisioned as traveling through space with a velocity, **u**, that can be decomposed into three Cartesian components: $\mathbf{u}_x$, $\mathbf{u}_y$, $\mathbf{u}_z$.



Cartesian components of velocity
The particle velocity **u** can be decomposed into velocity components along three Cartesian dimensions: $\mathbf{u}_x$, $\mathbf{u}_y$, $\mathbf{u}_z$.

The square of the velocity ($\mathbf{u}^2$) in terms of the three velocity components is

$$u^2 = u \cdot u = u_x^2 + u_y^2 + u_z^2$$

If the particle motion is random, the average velocities along all three Cartesian dimensions are equivalent.

Therefore, $\overline{u_x^2} = \overline{u_y^2} = \overline{u_z^2} = \frac{1}{3}\overline{u^2}$

$$P = \frac{1}{3}\frac{N}{V}m\overline{u^2}$$ -- eqn 3

n (mol) x M (g//mol) = m (g)

- Eqn 3 can be rewritten as $$PV = \frac{1}{3}nM\overline{u^2}$$

  where n is the amount in moles and M is the molar mass.

- By comparing this result to the ideal gas equation, PV = nRT, it can be concluded that $RT = \frac{1}{3}M\overline{u^2}$, and thus $\overline{u^2} = \frac{3RT}{M}$.

- $\therefore$ the root-mean-square speed, u_{rms}, is defined as

$$u_{rms} = \sqrt{\overline{u^2}} = \sqrt{\frac{\boldsymbol{3RT}}{\boldsymbol{M}}}$$ -- eqn 1

$$u_{rms} = \sqrt{\overline{u^2}} = \sqrt{\frac{3RT}{M}}$$ -- eqn 1

R = 8.3145 J·mol^{-1}·K^{-1}
J = kg·m^2·s^{-2}
M in kg·mol^{-1}

(see Class Practice 13)

$\because u_{rms} \propto \sqrt{T} \propto 1/\sqrt{M}$

$\therefore$ On average,

(1) molecular speeds ↑ as T ↑,

(2) speeds of lighter molecules > speeds of heavier molecules.

This result is based on a model (5 assumptions).
Is this valid?

Distribution of Molecular Speeds

- The molecules in a gas travel at different speeds. Because *N* is too large, we cannot know the speed of each molecule.
- A **statistical prediction** of how many molecules have a particular speed *u* can be made.
- The fraction, *F*, of molecules that have speed *u* is given by the following equation called ***Maxwell distribution of speeds (or Maxwell-Boltzmann distribution)***:

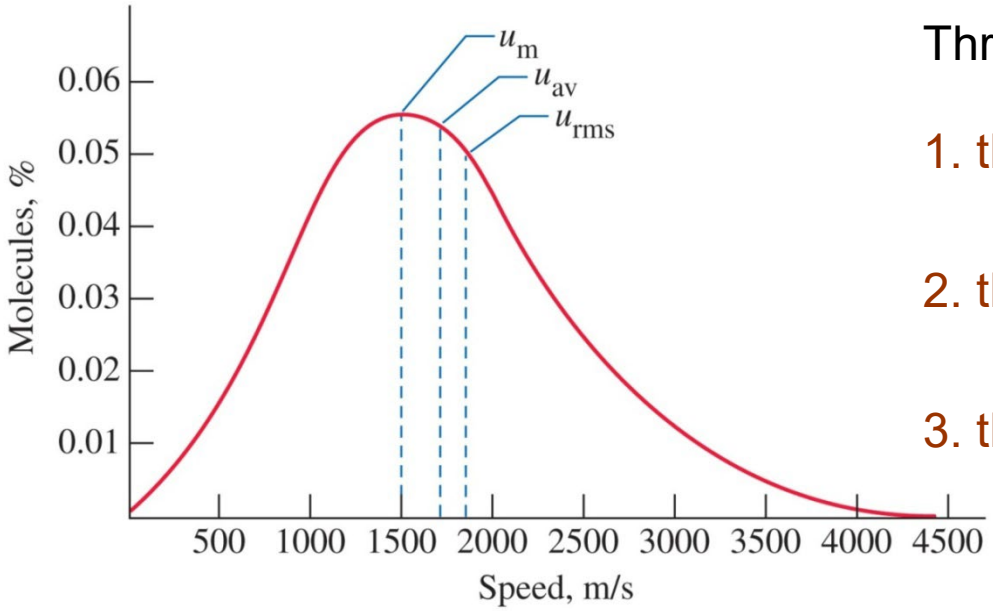
$$F(u) = 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} u^2 e^{-\left(\frac{Mu^2}{2RT}\right)}$$

For a certain value of u (in m/s), the corresponding value of F represents the fraction of molecules having speed between u and (u+1) m/s.

- This equation can help us understand how the distribution of speeds depends on molar mass, *M*, and temperature, *T*.

$$F(u) = 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} u^2 e^{-\left(\frac{Mu^2}{2RT}\right)}$$

u_m is the modal speed; u_{av} is the average speed; u_{rms} is obtained from the average of u^2: i.e. $\sqrt{\overline{u^2}}$



The shape of the distribution depends on the product of two opposing factors:

(1) u^2, and (2) $e^{-\left(\frac{Mu^2}{2RT}\right)}$

Distribution of molecular speeds –

The percentages of molecules with a certain speed are plotted as a function of speed.

Three different speeds are noted.

1. the most probable speed is ~1500 m/s;

modal speed

2. the average speed is ~1700 m/s; and

3. the root-mean-square speed is ~1800 m/s.

u_{rms}, obtained from the average of u^2

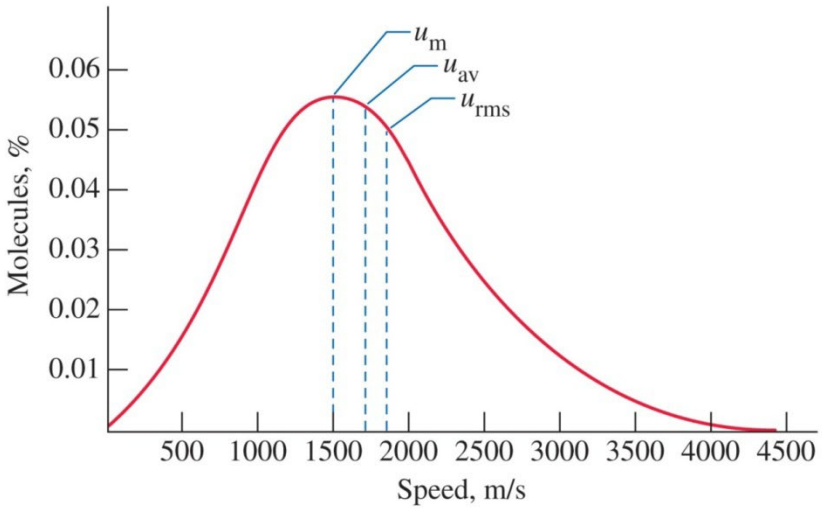
Notice that $u_m < u_{av} < u_{rms}$

For a certain value of u (in m/s), the corresponding value of F represents the fraction of molecules having speed between u and (u+1) m/s.

Look at the 3 characteristic speeds:

1. The most probable speed, or modal speed is denoted by u_m; more molecules have this speed than any other speed.
2. The average speed is denoted by u_{av}.
3. The root-mean-square speed, u_{rms}, is obtained from the average of u^2, which is denoted by $\overline{u^2}$.

Distribution of molecular speeds – hydrogen gas at 0°C



- These characteristic speeds can be calculated by using the following expressions:

$$u_m = \sqrt{\frac{2RT}{M}} \qquad u_{av} = \sqrt{\frac{8RT}{\pi M}} \qquad u_{rms} = \sqrt{\frac{3RT}{M}}$$ -- eqn 1

(R = 8.3145 J $K^{-1}mol^{-1}$)

It is the same as the derivation from the simple model.

Class Practice 13: Calculating a Root-Mean-Square Speed

EXAMPLE 6-14 Calculating a Root-Mean-Square Speed

Which is the greater speed, that of a bullet fired from a high-powered M-16 rifle (2180 mi/h) or the root-mean-square speed of H_2 molecules at 25 °C?

9.75×10^2 m/s

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

Class Practice 13: Calculating a Root-Mean-Square Speed

EXAMPLE 6-14 Calculating a Root-Mean-Square Speed

Which is the greater speed, that of a bullet fired from a high-powered M-16 rifle (2180 mi/h) or the root-mean-square speed of H_2 molecules at 25 °C?

9.75×10^2 m/s

$$u_{rms} = \sqrt{\overline{u^2}} = \sqrt{\frac{3RT}{M}}$$

Solution:

Analyze

This is a straightforward application of equation (6.19). We must use SI units: $R = 8.3145\ \text{J K}^{-1}\ \text{mol}^{-1}$ and $M = 2.016 \times 10^{-3}\ \text{kg mol}^{-1}$. Recall that $1\ \text{J} = 1\ \text{kg m}^2\ \text{s}^{-2}$.

Solve

Determine u_{rms} of H_2 with equation (6.19).

$$u_{\text{rms}} = \sqrt{\frac{3 \times 8.3145\ \text{kg m}^2\ \text{s}^{-2}\ \text{mol}^{-1}\ \text{K}^{-1} \times 298\ \text{K}}{2.016 \times 10^{-3}\ \text{kg mol}^{-1}}}$$

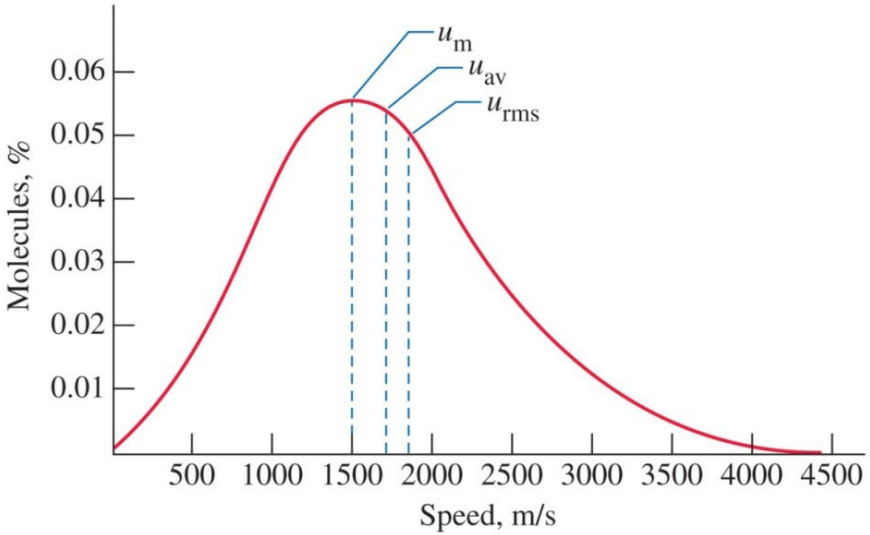
$$= \sqrt{3.69 \times 10^6\ \text{m}^2/\text{s}^2} = 1.92 \times 10^3\ \text{m/s}$$

The remainder of the problem requires us either to convert 1.92×10^3 m/s to a speed in miles per hour, or 2180 mi/h to meters per second. Then we can compare the two speeds. When we do this, we find that 1.92×10^3 m/s corresponds to 4.29×10^3 mi/h. The root-mean-square speed of H_2 molecules at 25 °C is greater than the speed of the high-powered rifle bullet.

Class Practice 13 (Con't)

Assess

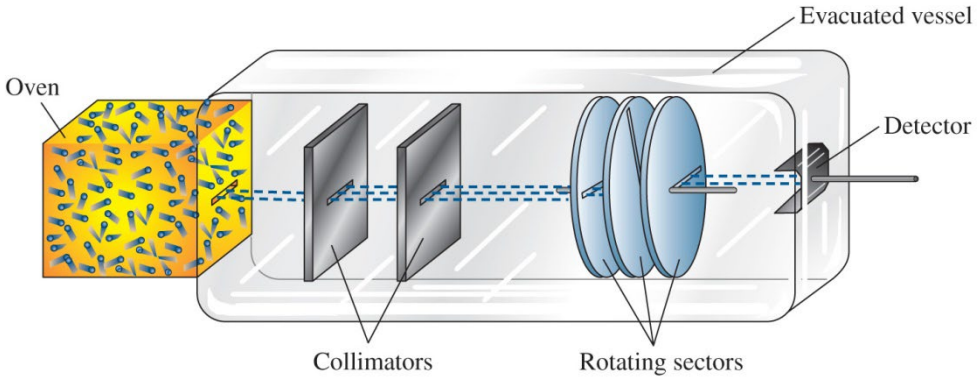
The cancellation of units yields a result for u_{rms} with the correct units (m/s). Also, Figure 6-15 shows that u_{rms} for H_2 is a bit greater than 1500 m/s at 273 K. At 298 K, u_{rms} should be slightly greater than it is at 273 K.

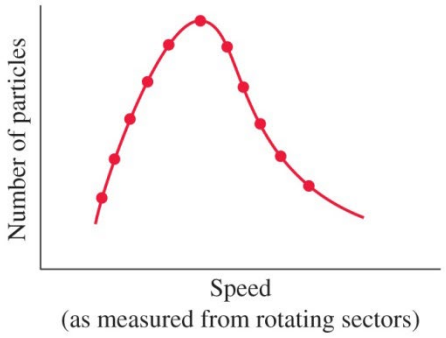


Distribution of molecular speeds – hydrogen gas at 0°C

Experimental Determination – Distribution of Molecular Speeds

Otto Stern's Experiment (1919)





- An oven (in which gas molecules are heated) and attached evacuated chamber are separated by a wall with a small hole in it. The gas molecules emerging from the hole pass through a series of slits, called collimators, as a beam. The beam then passes through a series of rotating disks, each disk has a slit cut in it.
- Only those molecules with the correct speed to pass through all rotating disks will reach the detector, where they can be counted.

For a given rotation speed, only those molecules with the right velocity can pass through the entire series of the disks. The rotation speeds of the disks are converted to molecular speeds.

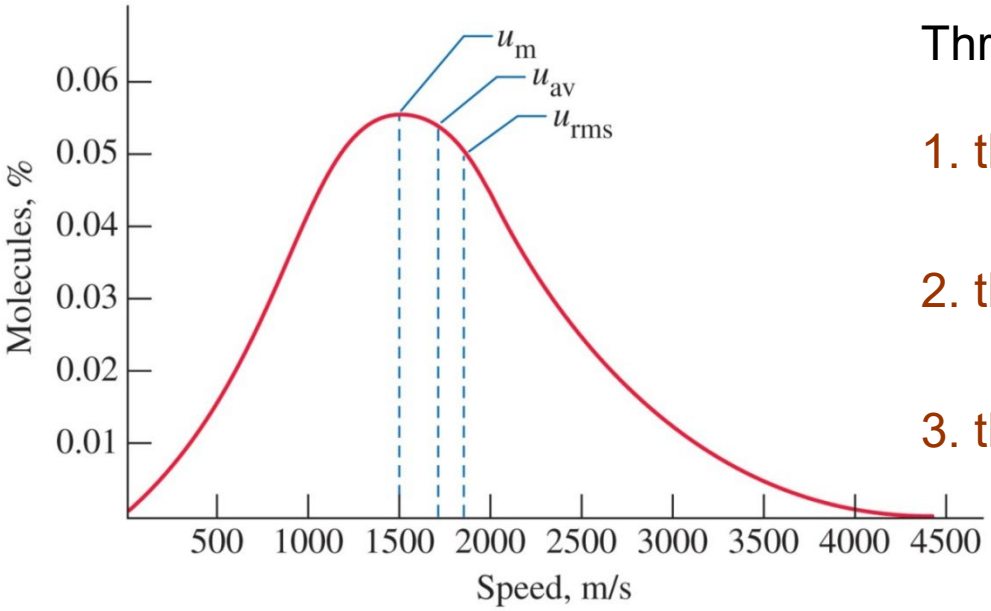
- By changing the rate of rotation of the disks, the complete distribution of molecular speeds can be determined.

This result is based on a model (5 assumptions).
Is this valid?

YES. It is experimentally proven and valid.

$$F(u) = 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} u^2 e^{-\left(\frac{Mu^2}{2RT}\right)}$$

u_m is the modal speed; u_{av} is the average speed; u_{rms} is obtained from the average of u^2: i.e. $\sqrt{\overline{u^2}}$



The shape of the distribution depends on the product of two opposing factors:

(1) u^2, and (2) $e^{-\left(\frac{Mu^2}{2RT}\right)}$

Distribution of molecular speeds – hydrogen gas at 0°C

The percentages of molecules with a certain speed are plotted as a function of speed.

Three different speeds are noted.

1. the most probable speed is ~1500 m/s;

modal speed

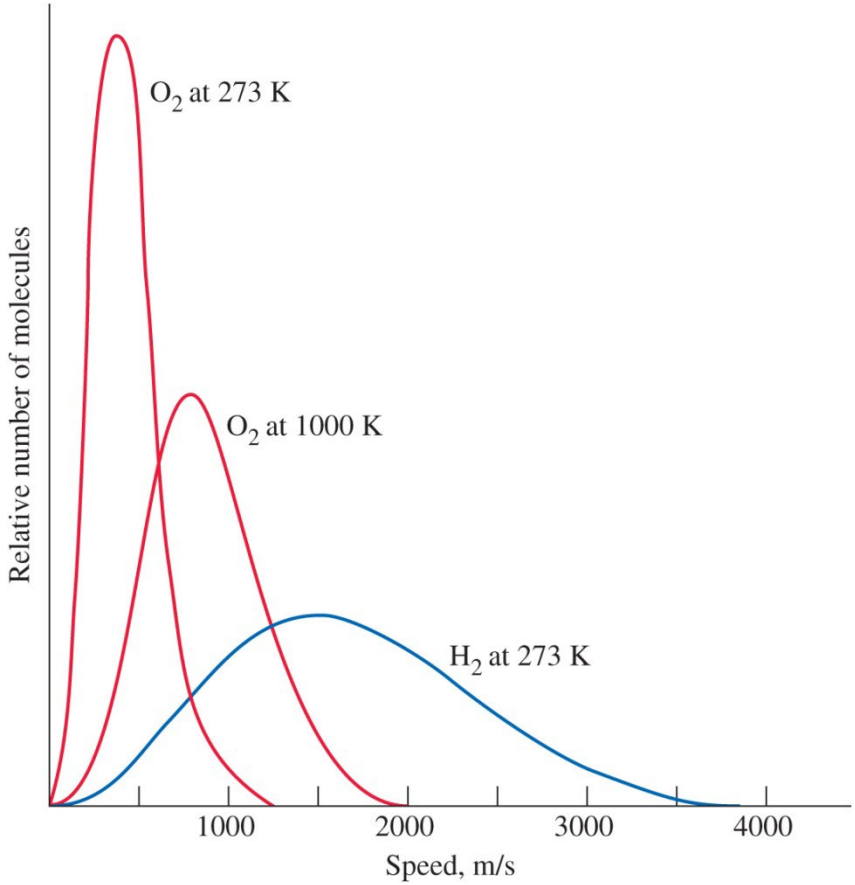
2. the average speed is ~1700 m/s; and

3. the root-mean-square speed is ~1800 m/s.

u_{rms}, obtained from the average of u^2

Notice that $u_m < u_{av} < u_{rms}$

$$F(u) = 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} u^2 e^{-\left(\frac{Mu^2}{2RT}\right)}$$



Distribution of Molecular Speeds – the effect of mass and temperature

The distribution depends on **temperature** and **molar mass**.

Look at the two curves for O_2:

As T ↑

- range of speeds broadens, and
- distribution shifts towards higher speeds,

Look at the curves for O_2 and H_2 at 273 K:

- The lighter the gas, the broader the range of speeds.

Note that the area under each curve is the same because N is the same

Effect of Temperature

- u_{rms} is of particular interest because we can use the value of u_{rms} to calculate the average kinetic energy, $\overline{E_k}$, of a collection of molecules.

- The average KE of a collection of molecules, each having a mass $m = M/N_A$, is the average of $\frac{1}{2}mu^2$:

$$\overline{E_k} = \frac{1}{2}m\overline{u^2} = \frac{1}{2} \times \frac{M}{N_A} \times (u_{rms})^2$$

- Since $u_{rms} = \sqrt{3RT/M}$, then $u^2_{rms} = \frac{3RT}{M}$, the expression above can be rewritten as:

$$\overline{E_k} = \frac{3}{2}\left(\frac{RT}{N_A}\right)$$ -- eqn 2

R and N_A are constant

That means, $\overline{E_k}$ = constant x T

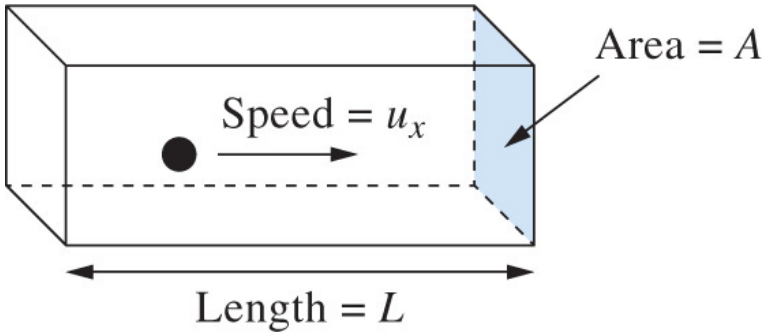
The **Kelvin temperature** (T) of a gas is **directly proportional** to the **average kinetic energy** ($\overline{E_k}$) of its molecules .

↑ freq ⇒ ↑ total force

1. The collision frequency – the no. of collisions per unit time.

2. The momentum transfer, or impulse per collision (momentum is transferred as the molecule reverses direction upon hitting the container wall). The momentum transfer to the wall is $+2mu_x$.

Why does PV=nRT applies to all gasses?

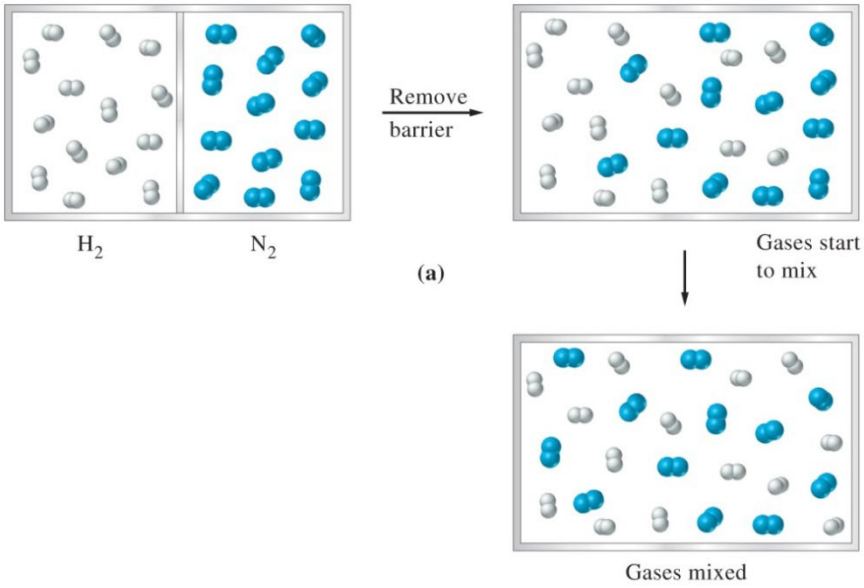


$$\overline{E_k} = \frac{1}{2}m\overline{u^2} \qquad \overline{E_k} = \frac{3}{2}\left(\frac{RT}{N_A}\right)$$

That means, $\overline{E_k}$ = constant x T

The **Kelvin temperature** (T) of a gas is **directly proportional** to the **average kinetic energy** ($\overline{E_k}$) of its molecules .

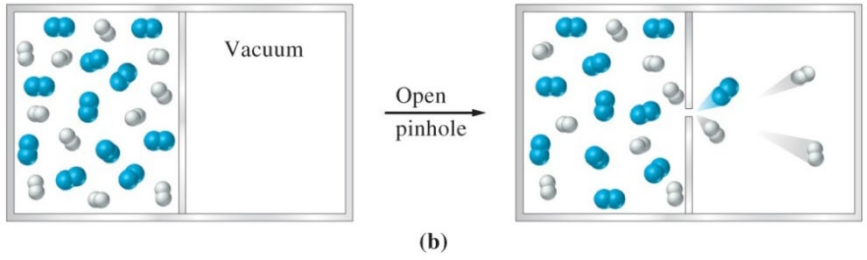
7. Gas Properties Relating to the Kinetic-Molecular Theory



(a) Diffusion

- is the migration of molecules as a result of random molecular motion.
- In this case, the H_2 initially diffuses farther through the N_2 because it is lighter. Eventually, a complete random mixing occurs (equilibrium).

$$u_{rms} = \sqrt{\overline{u^2}} = \sqrt{\frac{3RT}{M}}$$ -- eqn 1

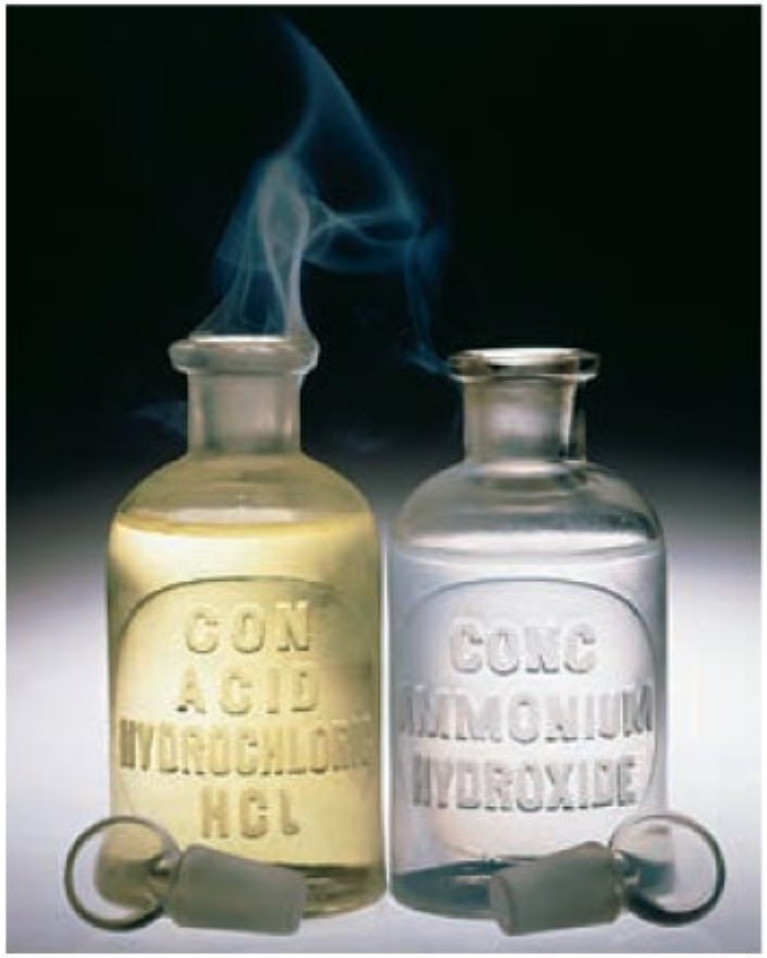


(b) Effusion

- a related phenomenon, is the escape of gas molecules from their container through a tiny pinhole into a vacuum (partial pressure).
- rate $\propto$ molecular speeds
- In this case, the lighter H_2 effuses faster (higher frequency) across the empty space than does the N_2.

(a) Diffusion

- is the migration of molecules as a result of random molecular motion.
- In this case, the H_2 initially diffuses farther through the N_2 because it is lighter. Eventually, a complete random mixing occurs (equilibrium).

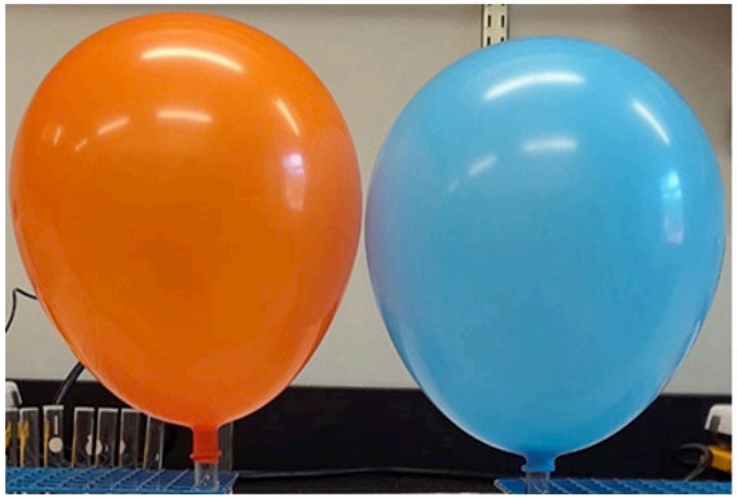


HCl (aq) NH_3 (aq)

$$NH_3\ (g) + HCl\ (g) \rightarrow NH_4Cl\ (s)$$

The gases diffuse toward each other and a white cloud of NH_4Cl forms (near the mouth of HCl(aq) container) as a result of the above reaction.

∵ NH_3 molecules (being lighter) diffuse faster than HCl molecules.





(b) Effusion

- a related phenomenon, is the escape of gas molecules from their container through a tiny pinhole into a vacuum (partial pressure).
- rate $\propto$ molecular speeds
- In this case, the lighter H_2 effuses faster (higher frequency) across the empty space than does the N_2.

Gas Properties Relating to the Kinetic-Molecular Theory

- Rate of effusion $\propto$ molecular speeds.

- Consider the effusion of two different gases at the same T & P, we can compare effusion rates with u_{rms}.

$$\frac{\text{rate of effusion of A}}{\text{rate of effusion of B}} = \frac{(u_{rms})_A}{(u_{rms})_B} = \sqrt{\frac{3RT/M_A}{3RT/M_B}} = \sqrt{\frac{M_B}{M_A}}$$

-- eqn 4

Refer to eqn 1

$$\frac{\text{rate of effusion of A}}{\text{rate of effusion of B}} = \frac{(u_{rms})_A}{(u_{rms})_B} = \sqrt{\frac{3RT/M_A}{3RT/M_B}} = \sqrt{\frac{M_B}{M_A}} \quad \text{-- eqn 4}$$

- Eqn 4 is a kinetic-theory statement of Graham's law.

Graham's law (applicable at low pressures only):
The rate of effusion of a gas is inversely proportional to the square root of its molar mass.

Limitations:
- Only for gases at ***low*** pressure (natural escape, not a jet)
- Tiny orifice (no collisions)
- Does **not** precisely apply to diffusion due to collisions

$\sqrt{\frac{M_B}{M_A}}$ Ratio can be used to obtain:

- Molecular speeds
- Effusion rates
- Effusion times
- Distances traveled by molecules
- Amounts of gas effused.



Class Practice 14

EXAMPLE 6-15 Comparing Amounts of Gases Effusing Through an Orifice

If 2.2×10^{-4} mol $N_2(g)$ effuses through a tiny hole in 105 s, then how much $H_2(g)$ would effuse through the same orifice in 105 s?

Class Practice 14

EXAMPLE 6-15 Comparing Amounts of Gases Effusing Through an Orifice

If 2.2×10^{-4} mol $N_2(g)$ effuses through a tiny hole in 105 s, then how much $H_2(g)$ would effuse through the same orifice in 105 s?

Solution:

Analyze

Let us reason qualitatively: H_2 molecules are lighter than N_2 molecules, so $H_2(g)$ should effuse faster than $N_2(g)$ when the gases are compared at the same temperature. Before we set the ratio

$$\frac{\text{mol } H_2 \text{ effused}}{\text{mol } N_2 \text{ effused}}$$

equal to $\sqrt{\text{ratio of molar masses}}$, we must ensure that the ratio of molar masses is *greater than 1*.

Solve

$$\frac{?\text{ mol } H_2}{2.2 \times 10^{-4} \text{ mol } N_2} = \sqrt{\frac{M_{N_2}}{M_{H_2}}} = \sqrt{\frac{28.014}{2.016}} = 3.728$$

$$?\text{ mol } H_2 = 3.728 \times 2.2 \times 10^{-4} = 8.2 \times 10^{-4} \text{ mol } H_2$$

Assess

We could have estimated the result before calculating it. Because the ratio of molar masses is approximately 14, the ratio of effusion rates is approximately $\sqrt{14}$, which is slightly smaller than 4. Therefore, H_2 will effuse almost 4 times as fast as N_2 and almost 4 times as much H_2 will effuse in the same period.



Class Practice 15

EXAMPLE 6-16 Relating Effusion Times and Molar Masses

A sample of Kr(g) escapes through a tiny hole in 87.3 s. The same amount of an unknown gas escapes in 42.9 s under identical conditions. What is the molar mass of the unknown gas?

Class Practice 15

EXAMPLE 6-16 Relating Effusion Times and Molar Masses

A sample of Kr(g) escapes through a tiny hole in 87.3 s. The same amount of an unknown gas escapes in 42.9 s under identical conditions. What is the molar mass of the unknown gas?

Solution:

Analyze

Because the unknown gas effuses faster, it must have a smaller molar mass than Kr. Before we set the ratio

$$\frac{\text{effusion time for unknown}}{\text{effusion time for Kr}}$$

equal to $\sqrt{\text{ratio of two molar masses}}$, we must make sure the ratio of molar masses is smaller than one. Thus, the ratio of molar masses must be written with the molar mass of the lighter gas (the unknown gas) in the numerator.

Solve

$$\frac{\text{effusion time for unknown}}{\text{effusion time for Kr}} = \frac{42.9\ \text{s}}{87.3\ \text{s}} = \sqrt{\frac{M_{\text{unk}}}{M_{\text{Kr}}}} = 0.491$$

$$M_{\text{unk}} = (0.491)^2 \times M_{\text{Kr}} = (0.491)^2 \times 83.80 = 20.2\ \text{g/mol}$$

Assess

Use the final result and work backward. The molar mass of the unknown is about 4 times as small as that of Kr. Because *effusion rate* $\propto 1/\sqrt{M}$, the unknown will effuse about $\sqrt{4} = 2$ times as fast as Kr. The effusion times show that the unknown does indeed effuse 2 times as fast as Kr.

An Ideal Gas is any gas that

(1) follows the ideal gas law

(2) has an internal energy that does not change if the temp of the gas does not change.

$$\left(\frac{\partial U}{\partial V}\right)_T = 0$$

$$\overline{E_k} = \frac{3}{2}\left(\frac{RT}{N_A}\right)$$

8. Non-ideal (Real) Gases

- Real gases are not 'ideal'. They show deviations from ideal gas law because molecules interact with one another and have molecular volumes.
- At or near ambient condition the deviations are marginal. However, at high pressures and low temperatures, the deviations become significant.
- For one mole of gas, if it is an ideal gas, $\frac{PV}{RT} = 1$
- For non-ideal gas, the quotient may be **greater/smaller** than 1.
- The quotient, $\frac{PV}{nRT}$, is called the ***compressibility factor***. It tells how much a gas deviates from ideal gas behaviour.

Densities and Molar Volumes of Various Gases

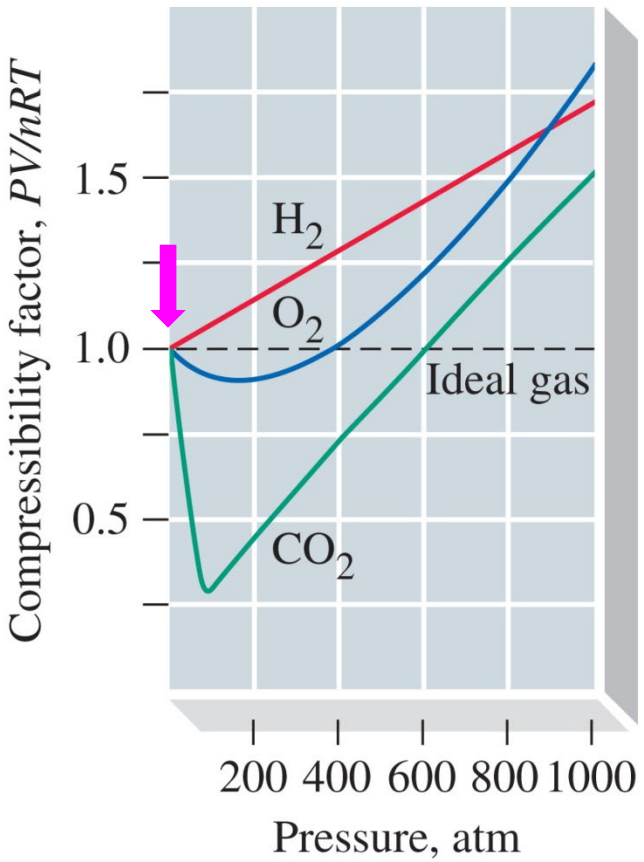
Gas	Molar Mass, g mol^{-1}	Density (at STP), g L^{-1}	Molar Volume,[a] L mol^{-1} (at STP) 0 °C, 1 bar	Molar Volume,[a] L mol^{-1} (at 0 °C, 1 atm)
H_2	2.01588	8.87104×10^{-2}	22.724	22.427
He	4.00260	0.17615	22.722	22.425
Ideal gas	–	–	22.711	22.414
N_2	28.0134	1.23404	22.701	22.404
CO	28.0101	1.23375	22.696	22.399
O_2	31.9988	1.41034	22.689	22.392
CH_4	16.0425	0.70808	22.656	22.360
NF_3	71.0019	3.14234	22.595	22.300
CO_2	44.0095	1.95096	22.558	22.263
N_2O	44.0128	1.95201	22.550	22.255
C_2H_6	30.0690	1.33740	22.483	22.189
NH_3	17.0352	0.76139	22.374	22.081
SF_6	146.0554	6.52800	22.374	22.081
C_3H_8	44.0956	1.98318	22.235	21.944
SO_2	64.064	2.89190	22.153	21.863

Source: All data are from the National Institute of Standards and Technology (NIST) *Chemistry WebBook*, available online at http://webbook.nist.gov/chemistry/.

[a]The molar volume is equal to the molar mass divided by the density. The molar volume at 0 °C and 1 atm is obtained by dividing the molar volume at STP by 1.01325.

The behaviour of real gases –

Compressibility factor as a function of pressure at 0°C



$$z = \frac{PV}{nRT}$$

Values of the compressibility factor **< 1**:

- signifies that intermolecular forces of **attraction** are largely responsible for deviations from ideal gas behaviour.

$$z = \frac{PV}{nRT}$$

Values of the compressibility factor **> 1**:

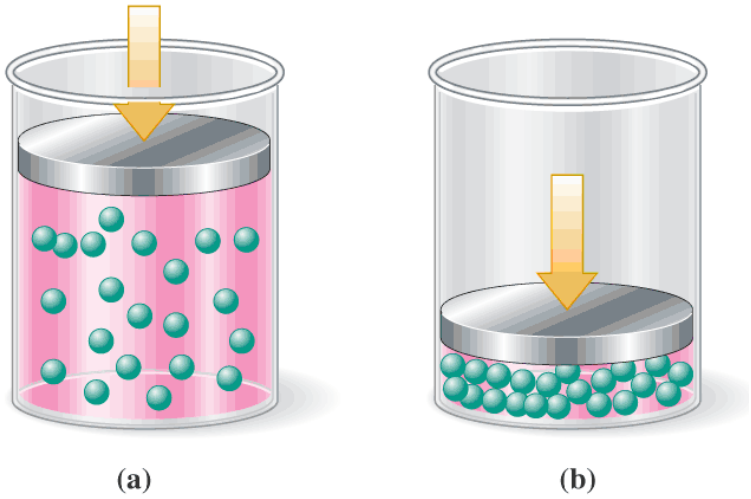
- when the **volume of the gas molecules** themselves is a significant fraction of the total gas volume.

All gases behave ideally at sufficiently low pressures (say below 1 atm).

↑ Pressure ⇒ ↑ deviation

Prediction from gas law:
At very high pressures, $V \rightarrow 0$
But actually this cannot be.

$\therefore$ At high P, $PV_{(actual)} > PV_{(predicted)}$,
$\Rightarrow$ compressibility ratio > 1

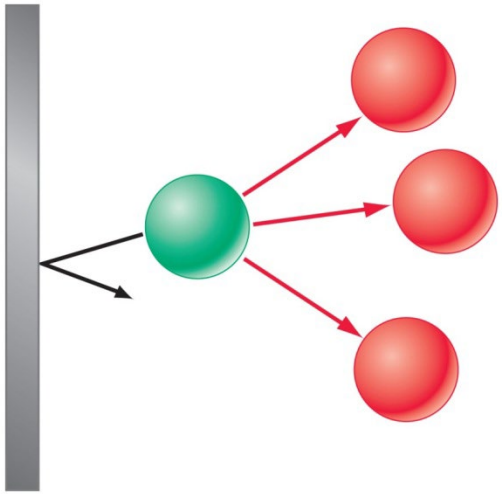


The effect of finite molecular size

Molecules do occupy space.

Kinetic Theory of Gases assumes
no intermolecular forces.
But actually there are.

$\therefore$ At low T, $PV_{(actual)} < PV_{(predicted)}$,
$\Rightarrow$ compressibility ratio < 1



Intermolecular forces of attraction

Attractive forces of the red molecules for the green molecule cause the green to exert less force when it collides with the wall than if these attractions did not exist.

Kinetic-Molecular Theory of Gases

- A gas is composed of a very large number of particles in constant, random, straight-line motion.
- Particles are point masses and separated by great distances.
- Collisions are rapid and elastic.

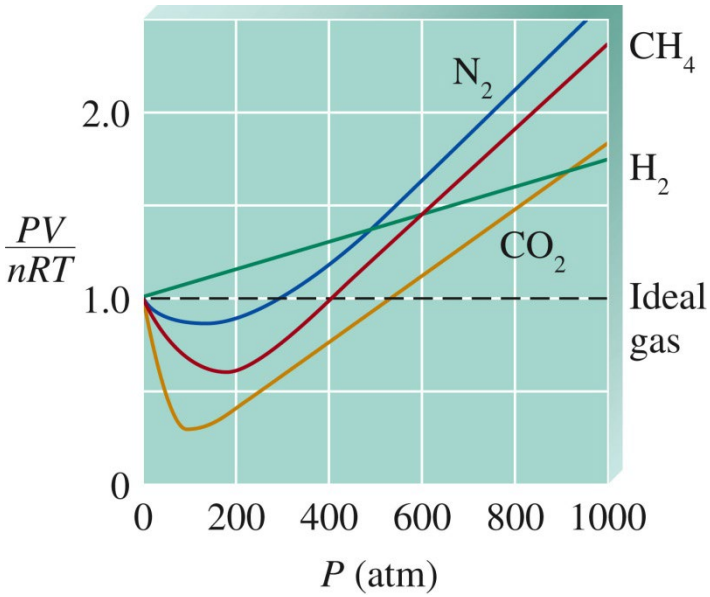
> An **elastic collision** is one that there is no loss of kinetic energy in the collision.

- No force between particles.
- Individual particles may gain or lose energy as a result of collisions. But total energy remains constant.

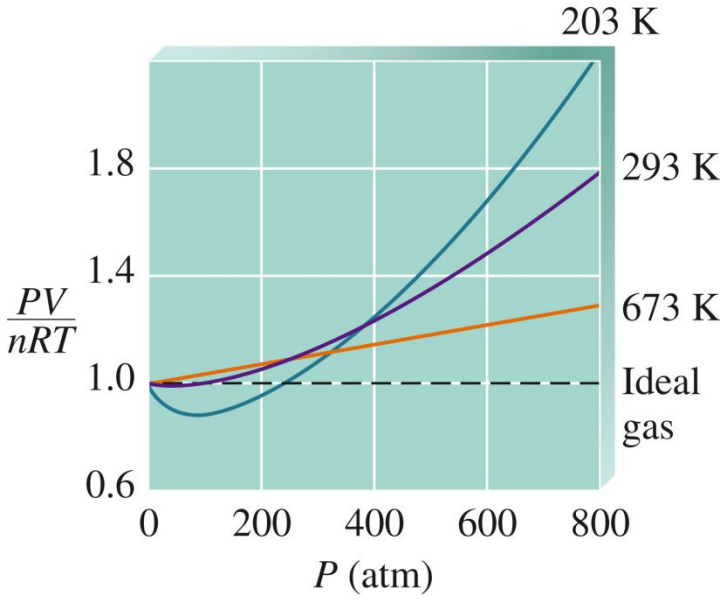


Visualizing Molecular Motion

The validity of this model can be checked by comparing predictions based on the model with experimental facts. It was proven that **the predictions based on this model are consistent with several observed macroscopic properties**.



Plots of *PV/nRT* versus *P* for Several Gases (200 K)



Plots of *PV/nRT* versus *P* for $N_2(g)$ at Three Temperatures

Gases tend to behave ideally at high temperatures and low pressures.

The van der Waals Equation

- A real gas equation of state, is defined with **van der Waals equation of state**.
- It takes both the finite size of molecules and the attractive forces into account.
- More accurate in calculating the relationship between *P*, *V*, and *T* for gases than the ideal gas law.

$$P = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$$

a is a measure of how strongly the molecules attract each other

b is called the excluded volume per mole (the volume that one mole of gas occupies when it condenses to a liquid).

- This equation of state has two parameters that are substance dependent and must be experimentally determined. The parameters *b* and *a, van der Waals constants,* take the finite size of the molecules and the strength of attractive interaction into account respectively.

The van der Waals Equation

Compressibility Factors and van der Waals Constants for Various Gases (at 10 bar and 300 K)

Gas	van der Waals Constants		Compressibility Factor
	a, bar L^2 mol^{-2}	b, L mol^{-1}	
H_2	0.2452	0.0265	1.006
He	0.0346	0.0238	1.005
Ideal gas	0	0	1
N_2	1.370	0.0387	0.998
CO	1.472	0.0395	0.997
O_2	1.382	0.0319	0.994
CH_4	2.303	0.0431	0.983
NF_3	3.58	0.0545	0.965
CO_2	3.658	0.0429	0.950
N_2O	3.852	0.0444	0.945
C_2H_6	5.580	0.0651	0.922
NH_3	4.225	0.0371	0.887
SF_6	5.580	0.0651	0.880
C_3H_8	9.39	0.0905	a
SO_2	7.857	0.0879	a

Source: van der Waals constants are from the *CRC Handbook of Chemistry and Physics*, 95th ed., David R. Lide (ed.)., Boca Raton, FL: Taylor & Francis Group, 2015. Compressibility factors are calculated by using data from the National Institute of Standards and Technology (NIST) Chemistry WebBook, available online at http://webbook.nist.gov/chemistry/.

[a]At 10 bar and 300 K, C_3H_8 and SO_2 are liquids.

The values of both ***a*** and ***b*** ↑ as the molecular sizes ↑.

The **smaller** the values of ***a*** and ***b***, the more closely the gas resembles an ideal gas.

Non-ideal (Real) Gases

Compressibility Factors and van der Waals Constants for Various Gases (at 10 bar and 300 K)

Gas	van der Waals Constants		Compressibility Factor
	a, bar L^2 mol^{-2}	b, L mol^{-1}	
H_2	0.2452	0.0265	1.006
He	0.0346	0.0238	1.005
Ideal gas	0	0	1
N_2	1.370	0.0387	0.998
CO	1.472	0.0395	0.997
O_2	1.382	0.0319	0.994
CH_4	2.303	0.0431	0.983
NF_3	3.58	0.0545	0.965
CO_2	3.658	0.0429	0.950
N_2O	3.852	0.0444	0.945
C_2H_6	5.580	0.0651	0.922
NH_3	4.225	0.0371	0.887
SF_6	5.580	0.0651	0.880
C_3H_8	9.39	0.0905	a
SO_2	7.857	0.0879	a

Source: van der Waals constants are from the *CRC Handbook of Chemistry and Physics*, 95th ed., David R. Lide (ed.)., Boca Raton, FL: Taylor & Francis Group, 2015. Compressibility factors are calculated by using data from the National Institute of Standards and Technology (NIST) Chemistry WebBook, available online at http://webbook.nist.gov/chemistry/.

[a]At 10 bar and 300 K, C_3H_8 and SO_2 are liquids.

H_2, He, N_2, CO, and O_2 (at 10 bar and 300 K) behave almost ideally ($PV/nRT \approx 1$)

NH_3, and SF_6 do not behave ideally (at 10 bar and 300 K).



Class Practice 16

EXAMPLE 6-17 Using the van der Waals Equation to Calculate the Pressure of a Gas

Use the van der Waals equation to calculate the pressure exerted by 1.00 mol $Cl_2(g)$ confined to a volume of 2.00 L at 273 K. For Cl_2, $a = 6.34\ \text{bar L}^2\ \text{mol}^{-2}$ and $b = 0.0542\ \text{L mol}^{-1}$.

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$



Class Practice 16

EXAMPLE 6-17 Using the van der Waals Equation to Calculate the Pressure of a Gas

Use the van der Waals equation to calculate the pressure exerted by 1.00 mol $Cl_2(g)$ confined to a volume of 2.00 L at 273 K. For Cl_2, $a = 6.34\ \text{bar L}^2\ \text{mol}^{-2}$ and $b = 0.0542\ \text{L mol}^{-1}$.

Solution:

Analyze

This is a straightforward application of equation (6.26). It is important to include units to make sure the units cancel out properly.

Solve

Solve equation (6.26) for P.

$$P = \frac{nRT}{V - nb} - \frac{n^2a}{V^2}$$

Then substitute the following values into the equation.

$$n = 1.00\ \text{mol}; V = 2.00\ \text{L}; T = 273\ \text{K}; R = 0.08314\ \text{bar L mol}^{-1}\ \text{K}^{-1}$$

$$n^2a = (1.00)^2\ \text{mol}^2 \times 6.34\ \frac{\text{bar L}^2}{\text{mol}^2} = 6.34\ \text{bar L}^2$$

$$nb = 1.00\ \text{mol} \times 0.0542\ \text{L mol}^{-1} = 0.0542\ \text{L}$$

$$P = \frac{1.00\ \text{mol} \times 0.08314\ \text{bar L mol}^{-1}\ \text{K}^{-1} \times 273\ \text{K}}{(2.00 - 0.0542)\ \text{L}} - \frac{6.34\ \text{bar L}^2}{(2.00)^2\ \text{L}^2}$$

$$P = 11.7\ \text{bar} - 1.6\ \text{bar} = 10.1\ \text{bar}$$



Class Practice 16 (Con't)

Assess

The pressure calculated with the ideal gas equation is 11.3 bar. By including only the b term in the van der Waals equation, we get a value of 11.7 bar. Including the a term reduces the calculated pressure by about 1.6 bar. Under the conditions of this problem, intermolecular forces of attraction are the main cause of the departure from ideal behavior. Although the deviation from ideality here is rather large, in problem-solving situations, you can generally assume that the ideal gas equation will give satisfactory results.